

BRAIN TUMOR CLASSIFICATION USING RESNET, VISION TRANSFORMER AND HYBRID RESNET - VIT

Project Report

*Submitted to the APJ Abdul Kalam Technological University in Partial
fulfilment of requirement for the award of degree*

Bachelor of Technology

In

Computer Science

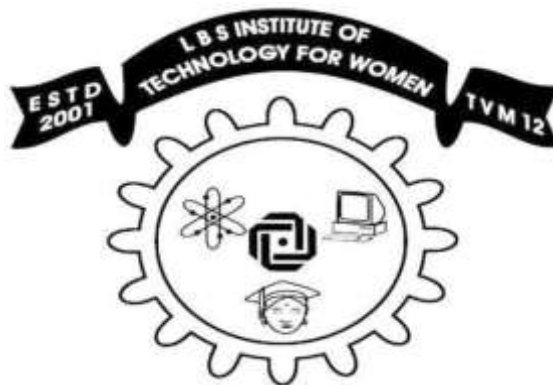
By

NANDANA N(LBT22CS077)

PANCHAMI M(LBT22CS091)

REKHA B(LBT22CS095)

VARADA A S(LBT22CS127)

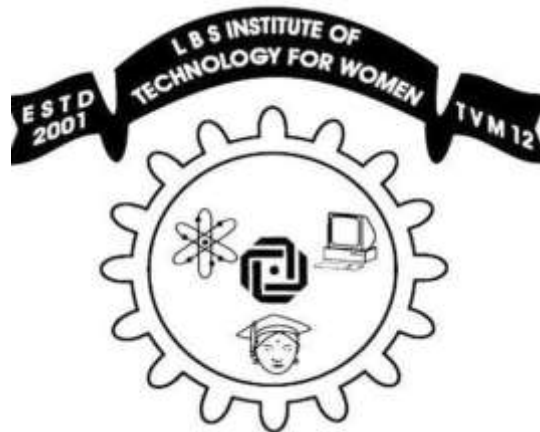


DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

LBS INSTITUTE OF TECHNOLOGY FOR WOMENS

TRIVANDRUM, KERALA APRIL 2026

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
LBS INSTITUTE OF TECHNOLOGY FOR WOMEN
TRIVANDRUM, KERALA



CERTIFICATE

This is to certify that the report entitled “**BRAIN TUMOR CLASSIFICATION USING RESNET, VISION TRANSFORMER AND HYBRID RESNET-VIT**” submitted by **NANDANA N[LBT22CS077], REKHA B[LBT22CS095], PANCHAMI M[LBT22CS091], VARADA A S[LBT22CS127]** to the APJ Abdul Kalam Technological University in partial fulfilment of the award of the requirement for the award of degree of Bachelor of Technology in Computer Science and Engineering is a bonafide record of the project carried out by them under our guidance and supervision.

Prof Aarathy S S
Assistant Professor
Department of CSE
LBSITW,TVM
(Project guide)

Dr Sumithra M D
Professor
Department of CSE
LBSITW,TVM
(Project coordinator)

Dr Sumithra M D
Professor
Department of CSE
Head of Department
LBSITW,TVM

DECLARATION

We undersigned hereby declare that the project report **BRAIN TUMOR CLASSIFICATION USING RESNET, VISION TRANSFORMER AND HYBRID RESNET - VIT** submitted for fulfilment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a Bonafide work done by us under the supervision of Prof. Aarathy S S, Assistant Professor, Department of Computer Science Engineering, LBS Institute of Technology for Women, Poojappura. This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misinterpreted or fabricated any data or idea or fact or source in my submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not previously formed the basis for the award of any degree, diploma, or similar title of any other University.

Place:Thiruvananthapuram

Date:April 2026

NANDANA N

REKHA B

PANCHAMI M

VARADA A S

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to all the people who have guided and assisted us through the course of this work. First and foremost, we take this opportunity to express our deep sense of gratitude and sincere thanks to our Principal, **Dr. Smithamol M B**, for rendering facilities within the campus for the completion of our mini project. We are greatly thankful to **Dr. Sumithra M D**, Head of the Department, Computer Science and Engineering for her support and cooperation. We are deeply indebted to our project guide **Prof. Aarathy S S**, Assistant Professor, Department of Computer Science and Engineering for his exceptional ability in providing positive criticism. We also take this opportunity to express our immense gratitude to **Prof. Biji S L**, Assistant Professor, Department of Computer Science and Engineering who put in their invaluable time and effort in keeping us aware of the different schedules of the project and evaluating our project with fair judgement. We thank the Department of Computer Science and Engineering of LBS Institute of Technology for Women, for giving us this opportunity to explore ourselves in the world of technology and present whatever we applied as a result. Finally, we thank our parents and near and dear ones who directly and indirectly contributed to the successful completion of our project.

NANDANAN

REKHA B

PANCHAMI M

VARADA A S

TABLE OF CONTENTS

TITLE	PAGE NO
ABSTRACT	I
LIST OF ABBREVIATIONS	II
LIST OF FIGURES	III-IV
LIST OF TABLES	V
CHAPTER 1:INTRODUCTION	1
1.1 BACKGROUND	1
1.2 CHALLENGES IN MEDICAL IMAGING	1
1.3 ARCHITECTURES UNDER STUDY	2
1.4 OBJECTIVES	2
1.5 PROPOSED SYSTEM	2
CHAPTER 2:LITERATURE REVIEW	4-24
CHAPTER 3:RESNET	25
3.1 INTRODUCTION TO RESNET	25
3.2 MOTIVATION FOR USING RESNET MODEL	25
3.3 ARCHITECTUREOF RESNET50	26
3.4 TRAINING METHODOLOGY	29
3.5 SUMMARY	31
CHAPTER 4: VISION TRANSFORMERS	33
4.1 INTRODUCTION TO VISION TRANSFORMERS	33
4.2 ARCHITECTURE OF VISION TRANSFORMERS	34
4.3 IMPLEMENTATION AND TRAINING METHODOLOGY	37
4.4 SUMMARY	39

CHAPTER 5:HYBRID RESNET-VISION TRANSFORMER	41
5.1 OVERVIEW OF THE HYBRID MODEL	41
5.2 MOTIVATION FOR USING A HYBRID ARCHITECTURE	41
5.3 ARCHITECTURE OF THE HYBRID RESNET-VIT	42
5.4 ADVANTAGES OF THE HYBRID MODEL	43
5.5 TRAINING METHODOLOGY	44
5.6 PERFORMANCE EVALUATION	45
5.7 SUMMARY OF HYBRID MODEL PERFORMANCE	46
CHAPTER 6:RESULT AND DISCUSSION	48
6.1 CLASSIFICATION RESULTS	48
6.2 COMPARATIVE ANALYSIS TRAINING PERFORMANCE	50
6.3 CONFUSION MATRIX ANALYSIS	56
6.4 PROTOTYPE AND DEPLOYMENT RESULTS	59
6.5 DISCUSSION	64
6.6 SUMMARY	65
CHAPTER 7:CONCLUSION	66
7.1 CONCLUSION	66
7.2 FUTURE SCOPE	67
REFERENCES	68

ABSTRACT

Medical imaging is a cornerstone of modern healthcare, providing essential data for the diagnosis and monitoring of complex diseases. With the rise of deep learning, automated analysis of Magnetic Resonance Imaging (MRI) has shown great potential in aiding clinical decisions. Historically, Convolutional Neural Networks (CNNs), such as ResNet, have been the primary choice for these tasks due to their efficiency in extracting local features like edges and textures. However, CNNs are limited in their ability to model global context and long range spatial relationships within an image. To overcome these challenges, Vision Transformers (ViTs) have recently been introduced, utilizing self-attention mechanisms to capture the global structure of an image. This project proposes a comparative study of three architectural approaches: ResNet, Vision Transformer (ViT), and a Hybrid ResNet-ViT model. The goal is to evaluate how these different paradigms perform in classifying brain tumors into various categories. By comparing a local-feature extractor (ResNet), a global-context model (ViT), and hybrid framework that combines both, this research aims to identify the most robust architecture for high-precision medical image classification.

LIST OF ABBREVIATIONS

ABBREVIATIONS	EXPANSION
CNN	Convolutional Neural Network
ViT	Vision Transformer
ResNet	Residual Network
AI	Artificial Intelligence
DL	Deep Learning
MRI	Magnetic Resonance Imaging
FLAIR	Fluid Attenuated Inversion Recovery
YOLO	You Only Look Once
XAI	Explainable Artificial Intelligence
NLP	Natural Language Processing
FFN	Feed Forward Network
MLP	Multi-Layer Perceptron
PE	Positional Encoding
mAP	Mean Average Precision
IoU	Intersection over Union
DSC	Dice Similarity Coefficient

LIST OF FIGURES

Figure No	Description	Page No
Fig no 3.1	ResNet50 Architecture	26
Fig no 4.1	ViT Architecture	34
Fig no 5.1	Hybrid Architecture	42
Fig no 6.1	Training and Validation Accuracy curves for ResNet50	51
Fig no 6.2	Training and Validation Loss curves for ResNet50	52
Fig no 6.3	Training and Validation Accuracy curves for ViT	53
Fig no 6.4	Training and Validation Loss curves for ViT	53
Fig no 6.5	Training and Validation Accuracy curves for Hybrid ResNet-ViT	54
Fig no 6.6	Training and Validation Loss curves for Hybrid ResNet-ViT	55
Fig no 6.7	Confusion Matrix for ResNet50	56
Fig no 6.8	Confusion Matrix for ViT	57
Fig no 6.9	Confusion Matrix for Hybrid ResNet-ViT	58
Fig no 6.10	Web application interface for deep learning model selection	60
Fig no 6.11	Web application interface for MRI scan upload and analysis	60
Fig no 6.12	Interface showing MRI scan with ResNet50 selected	61
Fig no 6.13	Diagnostic result displaying ResNet50 prediction	61
Fig no 6.14	Interface showing MRI scan with ViT selected	62
Fig no 6.15	Diagnostic result displaying ViT prediction	62
Fig no 6.16	Interface showing MRI scan with Hybrid ResNet50+ViT selected	63

Figure No	Description	Page No
Fig no 6.17	Diagnostic result displaying Hybrid model prediction	63

LIST OF TABLES

TABLE NO.	TITLE	PAGE NO.
Table no 2.1	Literature Review	14
Table no 6.1	Performance Metric of Resnet50	48
Table no 6.2	Performnce Metric of Hybrid Resnet-VIT	50
Table no 6.3	Comparison Table of 3 Models	64

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

The human brain is the most complex organ in the body, and the occurrence of a tumor can have devastating effects on cognitive and physical functions. Magnetic Resonance Imaging (MRI) has emerged as the most powerful non-invasive tool for visualizing the internal structures of the brain. However, the manual interpretation of these scans by radiologists is a labor-intensive process that is susceptible to inter-observer variability and human fatigue. With the advent of Deep Learning, Computer-Aided Diagnosis (CAD) systems have become a vital area of research. These systems aim to provide a "second opinion" to clinicians, ensuring that diagnoses are both rapid and objective. The evolution of these models has moved from simple neural networks to deep Convolutional Neural Networks (CNNs), and most recently, to Transformer-based architectures. This project explores this technological progression by applying three distinct deep learning paradigms to the specific problem of brain tumor classification, aiming to find the most accurate and computationally efficient solution.

1.2 CHALLENGES IN MEDICAL IMAGING

The automation of brain tumor classification is fraught with technical difficulties that traditional algorithms often fail to overcome:

- **High Intra-class Variability:** Tumors of the same type (e.g., Gliomas) can appear very differently in terms of shape, size, and intensity across different patients.
- **Low Inter-class Dissimilarity:** Certain types of tumors, such as Meningiomas and Pituitary tumors, may share similar visual characteristics in their early stages, making them difficult to distinguish even for trained eyes.
- **Global Context vs. Local Detail:** A standard CNN focuses on local pixel groups (textures). However, diagnosing a brain tumor often requires understanding the

"global" symmetry of the brain—knowing where the tumor is located relative to the ventricles or the midline of the skull.

1.3 ARCHITECTURES UNDER STUDY

ResNet (Residual Networks):- ResNet introduced the concept of "skip connections," allowing the training of very deep networks without the vanishing gradient problem. It serves as our baseline for localized spatial feature extraction.

Vision Transformer (ViT):- The ViT discards convolutions entirely, treating image patches as tokens in a sequence. It uses self-attention to learn long-range dependencies, helping the model understand how distant pixels in an MRI scan relate to one another.

Hybrid ResNet-ViT:- The Hybrid model is the state-of-the-art approach proposed in this project. It uses a ResNet "backbone" to extract high-level feature maps (local context) and then feeds those maps into a Transformer encoder (global context). This dual-pathway approach is hypothesized to provide the highest diagnostic accuracy.

1.4 OBJECTIVES

1. Data Acquisition and Preprocessing: To curate a comprehensive dataset of MRI scans and apply normalization and augmentation techniques.
2. Model Implementation: To develop and train three distinct architectures: ResNet, Vision Transformer, and a Hybrid ResNet-ViT.
3. Comparative Analysis: To evaluate these models based on Accuracy, Precision, Recall, and F1-Score to determine the most effective architecture.
4. Web Deployment: To design and implement a clinical decision support interface using Streamlit, allowing users to upload MRI scans and receive real-time classification results from the trained models.

1.5 PROPOSED SYSTEM

The proposed system is an end-to-end deep learning-based framework for automated brain tumor classification using MRI images

The system begins with MRI image input provided by the user through a web-based interface. The uploaded image undergoes preprocessing steps such as resizing, normalization, and augmentation to ensure uniformity and improve model performance

The processed image is then passed to three different deep learning models: ResNet50, Vision Transformer (ViT), and Hybrid ResNet-ViT. These models are trained to classify MRI images into four categories: Glioma, Meningioma, Pituitary tumor, and No Tumor

ResNet50 extracts local spatial features using convolutional layers, while the Vision Transformer captures global contextual relationships using self-attention mechanisms. The Hybrid ResNet-ViT model combines both approaches by using ResNet as a feature extractor and a Transformer encoder for global feature learning

After processing, the system outputs the predicted tumor class along with confidence scores. Performance is evaluated using metrics such as accuracy, precision, recall, and F1-score to ensure reliability of predictions

The system is integrated with a Streamlit-based web application, allowing users to upload MRI scans, select models, and view results in real-time. This provides an interactive platform for comparative analysis of different architectures

Overall, the proposed system acts as a Computer-Aided Diagnosis (CAD) tool that assists medical professionals in faster and more accurate brain tumor detection, reducing manual effort and improving diagnostic efficiency

CHAPTER 2

LITERATURE REVIEW

[1] A. A. Asiri, A. Shaf, T. Ali, M. A. Pasha, M. Aamir, M. Irfan, S. Alqahtani, A. J. Alghamdi, A. H. Alghamdi, A. F. A. Alshamrani, M. Alelyani, and S. Alamri, “Advancing Brain Tumor Classification through Fine-Tuned Vision Transformers: A Comparative Study of Pre-Trained Models,” *IEEE Sensors Journal*, vol. 23, no. 18, pp. 7913–7935, Sep. 2023, doi: 10.3390/s23187913.

The paper “Advancing Brain Tumor Classification through Fine-Tuned Vision Transformers: A Comparative Study of Pre-Trained Models” presents a comprehensive study on applying Vision Transformer (ViT) models for brain tumor classification using MRI images. The authors investigate whether fine-tuned, pre-trained ViT models can outperform existing deep learning approaches in medical image analysis. The study evaluates five pre-trained ViT architectures: R50-ViT-L16, ViT-B16, ViT-L16, ViT-L32, and ViT-B32, using a fine-tuning strategy. A publicly available Kaggle brain tumor MRI dataset containing 5712 images is used, classified into four categories: Glioma, Meningioma, Pituitary tumor, and No tumor. Out of these, 4855 images are used for training and 857 for testing. Standard preprocessing techniques such as image resizing, label encoding, and data augmentation are applied to improve model generalization. The models are trained using the Adam optimizer, categorical cross-entropy loss, and evaluated over 10 epochs. Performance is assessed using accuracy, precision, recall, F1-score, confusion matrix, and loss curves. Among all models, ViT-B32 achieves the best performance with an accuracy of 98.24%, outperforming both CNN-based and other transformer-based approaches. The results demonstrate that ViT models, even without task-specific pretraining, are highly effective for brain tumor classification when fine-tuned properly. The study concludes that Vision Transformers are powerful and reliable tools for medical image classification, offering improved accuracy and efficiency. The findings establish a strong benchmark for future research and suggest that ViT-based models can support early and accurate brain tumor diagnosis.

[2]V. Soumya, B. Nitish Kumar, and K. Muthulakshmi,“Automated Brain Tumor Detection and Segmentation Using ResNet and YOLO,” International Research Journal of Engineering and Technology (IRJET), vol. 12, no. 3, pp. 1172–1177, Mar. 2025.

This paper presents an automated deep learning–based system for brain tumor detection and segmentation using MRI images. The proposed approach combines ResNet-50 and YOLO (You Only Look Once) to improve diagnostic accuracy, speed, and reliability while reducing dependence on manual analysis by radiologists. Brain tumor diagnosis using MRI scans is often time-consuming, subjective, and prone to human error due to tumor variability in size, shape, and location. To address these challenges, the authors propose a hybrid ResNet–YOLO architecture. ResNet-50, a deep convolutional neural network with residual connections, is used as a feature extractor to capture complex spatial and structural patterns from MRI images. YOLO, a real-time object detection model, is employed for tumor localization and segmentation, enabling fast and precise identification of tumor regions. The model is trained and evaluated using a publicly available MRI dataset containing 10,208 images, including different MRI modalities such as T1, T2, FLAIR, and T1-contrast scans. Extensive data preprocessing techniques such as normalization, skull stripping, resizing, and data augmentation are applied to enhance model generalization. A patient-level train–validation–test split (80/10/10) is used to prevent data leakage and ensure robust evaluation. Model performance is assessed using accuracy, precision, recall, F1-score, mean Average Precision (mAP), Intersection over Union (IoU), and Dice Similarity Coefficient (DSC). The proposed system achieves a mean Average Precision (mAP) of 0.947, an overall accuracy of approximately 93.9%, and strong precision–recall balance, demonstrating effective tumor detection and segmentation. The study concludes that the ResNet-YOLO hybrid model provides a highly accurate, efficient, and reliable solution for automated brain tumor diagnosis. The approach has strong potential for clinical decision support, enabling faster diagnosis, early intervention, and improved patient outcomes in neuro-oncology.

[3] D. Vamsidhar, P. Desai, S. Joshi, S. Kolhar, N. Deshpande, and S. Gite, “Hybrid model integration with explainable AI for brain tumor diagnosis: A unified approach to MRI analysis and prediction,” *Scientific Reports*, vol. 15, 2025, doi: 10.1038/s41598-025-06455-2.

The paper presented focuses on the application of deep learning techniques for automated brain tumor detection and classification using MRI images. Brain tumors are life-threatening conditions that require early and accurate diagnosis to improve patient survival rates. Traditional diagnosis methods depend heavily on manual interpretation by radiologists, which is time-consuming and prone to human error. To overcome these challenges, the study proposes an automated computer-aided diagnosis system. The proposed approach utilizes Convolutional Neural Networks (CNNs) to extract meaningful features from brain MRI scans. The system follows a structured pipeline that includes data collection, preprocessing, feature extraction, model training, and performance evaluation. Preprocessing steps such as image resizing, normalization, noise removal, and data augmentation are applied to improve model robustness and generalization. The deep learning model is trained on a labeled MRI dataset containing tumor and non-tumor images. The trained model learns to differentiate between healthy brain tissues and tumor-affected regions by analyzing spatial and intensity patterns in MRI scans. Performance evaluation is carried out using standard metrics such as accuracy, precision, recall, F1-score, and confusion matrix, which demonstrate the effectiveness of the proposed method. The experimental results indicate that the proposed deep learning-based system achieves high accuracy and reliable classification performance, making it suitable for clinical assistance. The study concludes that automated brain tumor detection using deep learning can significantly reduce diagnosis time, minimize human errors, and support radiologists in making faster and more accurate decisions. Overall, the paper highlights the potential of AI-based medical image analysis systems in enhancing diagnostic efficiency and improving patient care in neuro-oncology.

[4] A. J. Aiya, N. Wani, M. Ramani, A. Kumar, S. Pant, K. Kotecha, A. Kulkarni, and A. Al-Danakh, "Optimized deep learning for brain tumor detection: A hybrid approach with attention mechanisms and clinical explainability," *Scientific Reports*, vol. 15, no. 1, pp. 1–16, 2025, doi: 10.1038/s41598-025-04591-3.

This paper presents an optimized hybrid deep learning framework for brain tumor classification using MRI images, with a strong emphasis on clinical interpretability alongside high diagnostic accuracy. The study addresses a key limitation of conventional deep learning models in medical imaging—their black-box nature—by integrating attention mechanisms and explainable AI techniques. The proposed approach uses a VGG16-based transfer learning architecture enhanced with a SoftMax-weighted channel attention mechanism. This attention layer prioritizes diagnostically relevant features, allowing the model to focus on tumor-specific regions rather than treating all image regions equally. To further improve transparency, the model incorporates Grad-CAM visualizations, which generate heatmaps highlighting image regions that influence classification decisions. These explanations were validated by expert radiologists, increasing trust in clinical applicability. The model was trained and evaluated on a publicly available Kaggle brain MRI dataset, covering four classes: glioma, meningioma, pituitary tumor, and no tumor. Extensive data augmentation, hyperparameter tuning (using Keras Tuner), and regularization techniques such as dropout and early stopping were applied to enhance generalization and prevent overfitting. Training was conducted over 20 epochs using the Adam optimizer and categorical cross-entropy loss. Experimental results demonstrate that the proposed hybrid model achieves high classification accuracy, precision, recall, and F1-scores, outperforming several existing CNN-based and ensemble methods on the same dataset. The attention-enhanced architecture improves feature discrimination, while Grad-CAM provides meaningful visual explanations aligned with clinical reasoning. In conclusion, the paper introduces a clinically reliable and interpretable brain tumor classification system that balances performance and explainability. The authors highlight the model's potential for real-world diagnostic support and suggest future work involving larger datasets, improved generalization, and broader clinical validation.

[6] K. M. Hosny and M. A. Mohammed, “Explainable AI and vision transformers for detection and classification of brain tumor: A comprehensive survey,” *Artificial Intelligence Review*, vol. 58, p. 259, Jun. 2025, doi: 10.1007/s10462-025-11221-x.

The paper titled “*Explainable AI and Vision Transformers for Detection and Classification of Brain Tumor: A Comprehensive Survey*” presents an extensive review of recent deep learning techniques used for brain tumor detection and classification from MRI images. The authors analyze studies published between 2020 and 2024, focusing on convolutional neural networks

(CNNs), transfer learning, vision transformers (ViTs), hybrid models, and explainable artificial intelligence (XAI) methods. The survey highlights commonly used datasets such as BraTS and Figshare, along with preprocessing techniques and evaluation metrics. Special emphasis is placed on explainable AI methods like Grad-CAM, LIME, and SHAP to address the lack of interpretability in deep learning models, which is critical for clinical trust. The paper also discusses major challenges in brain tumor diagnosis, including data scarcity, class imbalance, computational complexity, and the black-box nature of deep models. Finally, it outlines future research directions, recommending hybrid CNN–ViT architectures, improved interpretability, and data augmentation techniques to enhance reliability and clinical applicability of AI-based brain tumor diagnostic systems

[7] F. J. Dorfner, J. B. Patel, J. Kalpathy-Cramer, E. R. Gerstner, and C. P. Bridge, “A review of deep learning for brain tumor analysis in MRI,” *npj Precision Oncology*, vol. 9, no. 2, pp. 1–?, 2025. doi: 10.1038/s41698-024-00789-2.

The paper “*A Review of Deep Learning for Brain Tumor Analysis in MRI*” explores the rapid progress of deep learning (DL) in neuro-oncology, focusing specifically on MRI-based brain tumor analysis. It explains how DL has enabled automated tumor segmentation, classification, and quantitative assessment, improving diagnostic accuracy, treatment planning, and disease monitoring. The review begins by describing the evolution from handcrafted radiomic feature approaches to modern CNN-based deep learning models capable of learning patterns directly from imaging data. A major portion

discusses segmentation, the most developed application, detailing influential architectures such as U-Net, its modifications, nnU-Net, cascaded networks, and ensemble approaches that have consistently led benchmarks like the BraTS challenge. The authors also highlight public datasets—including BraTS, TCGA, and others—that have accelerated research but also note variability in data quality, modalities, and annotations that complicate model generalization. Beyond segmentation, the paper evaluates classification tasks such as distinguishing recurrence from radiation necrosis, predicting overall survival, and inferring genetic biomarkers like IDH mutation or MGMT methylation status directly from MRI. While promising models exist, the authors emphasize that biomarker prediction remains challenging, with mixed results and concerns regarding reproducibility. Quantification tools for tumor volume measurements and response assessment are discussed, showing that DL-based volumetric methods outperform traditional 2D RANO criteria. The paper also highlights challenges including domain shift across scanners, missing MRI sequences, pediatric tumor differences, uncertainty estimation, fairness and bias, reproducibility issues, and the need for clinical explainability. Looking forward, the authors discuss potential directions such as quantitative MRI, multimodal data integration, transformers, and foundation models trained on massive datasets, suggesting that future DL systems could enable more accurate, personalized treatment planning and may soon play a significant role in clinical workflows.

[8] R. A. Zeineldin *et al.*, “Explainable hybrid vision transformers and convolutional network for multimodal glioma segmentation in brain MRI,” *Scientific Reports*, vol. 14, no. 3713, 2024. doi: 10.1038/s41598-024-54186-7.

The paper introduces TransXAI, an explainable hybrid deep learning framework designed for automated glioma segmentation in brain MRI using multimodal data (T1, T1Gd, T2, FLAIR). The model combines Convolutional Neural Networks (CNNs) for extracting fine-grained local features and Vision Transformers (ViT) for capturing long-range dependencies, forming a robust encoder-decoder architecture that improves feature representation across tumor sub-regions. To address the “black-box” nature of deep learning models, the authors integrate an explainability module based on Grad-CAM, generating heatmaps that visually highlight tumor-relevant regions and demonstrate how

the network interprets input features. Trained on the BraTS 2019 dataset, TransXAI achieved competitive performance with Dice scores of 0.882 (WT), 0.782 (TC), and 0.745 (ET), along with strong generalization validated on external FeTS2022 data. The model not only shows high segmentation accuracy but also provides interpretable saliency maps, enabling clinicians to better understand model decisions and increasing trust in real clinical applications. Clinical expert evaluation confirmed the usefulness of visual explanations, indicating TransXAI's potential for integration in neurosurgical workflow, diagnosis support, and future extension to 3D MRI analysis.

[9] H. Yan, V. Mubonanyikuzo, T. E. Komolafe, L. Zhou, T. Wu, and N. Wang, "Hybrid-RViT: Hybridizing ResNet-50 and Vision Transformer for Enhanced Alzheimer's disease detection," *PLOS ONE*, vol. 20, no. 2, pp. 1–18, Feb. 2025. doi: 10.1371/journal.pone.0318998.

The research paper titled *"Hybrid-RViT: Hybridizing ResNet-50 and Vision Transformer for Enhanced Alzheimer's Disease Detection"* presents a deep learning-based hybrid model designed to improve the early detection and classification of Alzheimer's disease (AD) using brain MRI images. The study emphasizes the importance of early diagnosis, as AD is a progressive neurodegenerative disorder that severely affects memory and cognitive function, with increasing global prevalence. Traditional CNN-based models and standalone Vision Transformers (ViT) have shown promise in medical image analysis, but each has limitations—CNNs struggle with long-range spatial dependencies, while ViTs require large datasets. To overcome these challenges, the authors propose a combined approach leveraging the strengths of both architectures. The Hybrid-RViT model integrates ResNet-50 for local feature extraction and transfer learning with ViT for capturing global contextual information via self-attention. The model was trained on T1-weighted MRI scans from the OASIS dataset, containing subjects with different stages ranging from normal cognition to varying levels of dementia. Preprocessing included resizing, normalization, splitting the dataset into training/validation/testing sets, and using Nipype to prevent data leakage. Experimental results demonstrated strong performance, achieving 97% training accuracy and 95% test accuracy, outperforming existing CNN, ViT, and hybrid approaches. Evaluation metrics (precision, recall,

F1-score) further confirmed reliable classification across AD stages. Ablation studies also revealed that deeper backbones like ResNet-101 led to overfitting, validating ResNet-50 as a suitable choice. The proposed methodology shows

potential for clinical use, offering efficient AD classification from MRI images with high accuracy. The authors highlight its value in early diagnosis, treatment planning, and supporting healthcare decisions. However, they also acknowledge limitations such as dataset size, generalization to multi-center data, and computational demands of ViT. Future work aims to improve model robustness, incorporate multimodal data, expand datasets, and adapt the model for real-time or portable medical applications. Overall, the Hybrid-RViT model presents a significant step toward effective AI-assisted Alzheimer's disease detection.

[10] M. Agarwal, G. Rani, A. Kumar, P. Kumar K, R. Manikandan, and A. H. Gandomi, "Deep learning for enhanced brain tumor detection and classification," *Results in Engineering*, vol. 22, p. 102117, 2024.

The paper *"Deep Learning for Enhanced Brain Tumor Detection and Classification"* presents an automated hybrid system designed to improve the early detection and classification of brain tumors using MRI images. The study introduces a two-phase system named Auto Contrast Enhancer, Tumor Detector and Classifier, which aims to overcome challenges such as low contrast in MRI scans, manual diagnosis limitations, and high time consumption in the medical screening process. In the first phase, the authors employ the ODTWCHE technique, an optimized contrast enhancement method that enhances low-contrast MRI images using histogram segmentation, gamma correction, and Wiener filtering, resulting in clearer tumor visualization without losing essential image details. In the second phase, the system utilizes deep transfer learning, where multiple pre-trained CNN models including AlexNet, VGG-16, VGG-19, DenseNet-201, ResNet-50, and Inception V3 are compared for tumor classification, finally selecting Inception V3 due to its superior performance. The model automatically extracts high- and low-level features for binary classification of tumors into benign and malignant categories. Experimental evaluation was performed using the Figshare dataset,

consisting of 3064 training and 397 testing MRI images. The proposed system achieved an impressive 98.89% classification accuracy, outperforming existing models in the literature. The results also demonstrated high sensitivity for both benign and malignant tumor categories, proving the robustness of the hybrid model even on images with varying brightness and contrast levels. The work highlights the importance of combining contrast enhancement with deep learning architecture for reliable automated diagnosis. The authors conclude that the model shows significant potential for real-world clinical assistance and suggest improvements such as deployment on cloud-based platforms, addition of more diverse datasets, and integration of explainable AI for better decision support in medical practice.

[11]M. Ahmed, M. S. Miah, M. M. Rahman, M. Niu, and M. K. Islam, “Brain tumor detection and classification in MRI using hybrid ViT and GRU model with explainable AI in Southern Bangladesh,” *Scientific Reports*, vol. 14, no. 22797, 2024, doi: 10.1038/s41598-024-71893-3.

The paper provides an extensive review of how deep learning techniques, particularly convolutional neural networks (CNNs), are applied for automatic brain tumor detection and classification using MRI images. It emphasizes that traditional diagnosis of brain tumors heavily relies on expert radiologists, which is not only time-consuming but also susceptible to human interpretation errors, especially when dealing with large image volumes or poor-quality scans. Early identification of brain tumors is crucial, as timely detection significantly increases patient survival rates and can guide appropriate treatment planning. To address these challenges, the authors propose a computer-aided diagnosis (CAD) system powered by deep learning, aiming to automate the detection process and improve diagnostic reliability.

In the methodology, MRI images first undergo several preprocessing steps, including normalization, resizing, and noise reduction to enhance image clarity and prepare them for feature extraction. This capability makes CNNs highly effective for medical image analysis. The developed system is trained using labeled MRI data and is designed to classify images into tumor and non-tumor categories. During experimentation, the model demonstrated strong performance, achieving high accuracy in identifying tumors,

minimizing false predictions, and showing robustness in recognizing subtle tumor regions within MRI scans. The evaluation results clearly indicate that deep learning-based models offer superior performance compared to classical methods, which often fail to generalize across diverse image variations. The study concludes that the proposed CNN-driven model has great potential for integration into clinical workflows, offering rapid, automated, and reliable

tumor screening support. However, the authors also acknowledge certain limitations, such as dependency on dataset size, lack of multi-class tumor type classification, and the need for more advanced optimization for real-time medical use. Future research directions include expanding dataset diversity, incorporating multi-class tumor grading, and integrating explainable AI to improve transparency and clinical trust. Overall, the research highlights the effectiveness of deep learning in brain tumor classification and paves the way for further advancements in AI-assisted medical diagnostics.

SL NO	TITLE	METHODOLOGY	PROS	CONS
1	A. A. Asiri, A. Shaf, T. Ali, M. A. Pasha, M. Aamir, M. Irfan, S. Alqahtani, A. J. Alghamdi, A. H. Alghamdi, A. F.A. Alshamrani, M. Alelyani, and S. Alamri, “Advancing Brain Tumor Classification through Fine-Tuned Vision Transformers: A Comparative Study of Pre-Trained Models,” <i>IEEE Sensors Journal</i> , vol. 23, no. 18, pp. 7913–7935, Sep. 2023, doi: 10.3390/s23187913.	MRI preprocessing and data augmentation Fine-tuning pre-trained VisionTransformer Models: R50-ViT, ViT-B/L-16, ViT-B/L-32 Four-class brain tumor classification	High Classification Accuracy Use of Multiple Pre-trained Models Effective Fine-Tuning Strategy Robust Evaluation Metrics Handles Multiple Tumor Classes Outperforms Existing Method	High Computational Cost Limited Dataset Diversity Risk of Overfitting Lack of Explainability No Cross-Dataset Validation Focus Only on Classification Deployment Aspects Not Discussed

SL NO	TITLE	METHODOLOGY	PROS	CONS
2	V. Soumya, B. Nitish Kumar, and K. Muthulakshmi, "Automated Brain Tumor Detection and Segmentation Using ResNet and YOLO," International Research Journal of Engineering and Technology (IRJET), vol. 12, no. 3, pp. 1172–1177, Mar. 2025.	Collected labeled brain MRI images from a public dataset Applied preprocessing (denoising, normalization, resizing, skull stripping) Performed data augmentation to improve generalization. Split data using patient-level 80/10/10 train–validation–test ratio. Used ResNet-50 as backbone for deep feature extraction Integrated YOLO for real-time tumor detection and segmentation Fine-tuned the hybrid ResNet–YOLO model on MRI images	Achieves high detection performance (mAP $\approx$ 0.947) Supports real-time tumor localization using YOLO Uses patient-level data split, reducing data leakage Applies extensive preprocessing and data augmentation, improving generalization. Evaluated with multiple metrics (accuracy, precision, recall, IoU, Dice) Suitable for clinical decision support	May struggle with very small tumors due to YOLO limitations Uses relatively limited dataset, affecting generalization Lacks external validation on independent datasets Requires higher computational resources Limited explainability (black-box model) Does not provide highly precise tumor boundary segmentation

SL NO	TITLE	METHODOLOGY	PROS	CONS
3	D.Vamsidhar, P. Desai, S. Joshi, S. Kolhar, N. Deshpande, and S. Gite, "Hybrid model integration with explainable AI for brain tumor diagnosis: A unified approach to MRI analysis and prediction," Scientific Reports, vol. 15, 2025, doi: 10.1038/s41598-025-06455-2.	MRI data collection Image preprocessing and augmentation CNN-based feature extraction Model training and classification Performance evaluation	Automated brain tumor detection Reduces manual diagnosis errors High classification accuracy Faster diagnosis process Supports radiologist decision-making Uses standard evaluation metrics	CNN-based model only Limited explainability mentioned Dataset dependency not discussed Real-time deployment not specified No tumor localization or segmentation

SL NO	TITLE	METHODOLOGY	PROS	CONS
4	A. J. Aiya, N. Wani, M. Ramani, A. Kumar, S. Pant, K. Kotecha, A. Kulkarni, and A. Al-Danakh, "Optimized deep learning for brain detection: tumor A hybrid approach with attention mechanisms and clinical explainability," Scientific Reports, vol. 15, no. 1, pp. 1–16, 2025, doi: 10.1038/s41598-025-04591-3.	Collected and reviewed brain tumor research papers (2020–2024). Analyzed deep learning models like CNNs, ViTs, and hybrid CNN–ViT methods. Studied commonly used MRI datasets (BraTS, Figshare). Reviewed preprocessing techniques (normalization, augmentation). Compared evaluation metrics (accuracy, precision, recall, F1-score). Examined explainable AI methods (Grad-CAM, LIME, SHAP). Identified challenges and future research directions.	High classification accuracy Uses explainable AI (Grad-CAM) Attention mechanism improves focus Reduces black-box behavior Clinically interpretable results Validated by radiologists	Based on VGG16 (older model) Uses only one public dataset High computational cost Limited real-world testing No tumor segmentation 2D MRI analysis only

SL NO	TITLE	METHODOLOGY	PROS	CONS
5	A. Karagoz, O. U. Nalbantoglu, and G. C. Fox, "Residual Vision Transformer (ResViT) based self-supervised learning model for brain tumor classification," arXiv preprint arXiv:2411.12874, Nov. 2024.	Uses a two-stage self-supervised learning framework. Pre-trains a ResViT model for MRI image synthesis without labels. Combines CNN (local features) and Vision Transformer (global features). Generates synthetic MRI images for data augmentation. Transfers pretrained weights to a ResViT-based classifier. Fine-tunes the model for brain tumor classification. Classifies MRI images into four tumor classes. Evaluated on BraTS, Figshare, and Kaggle datasets.	Uses self-supervised learning, reducing the need for large labeled MRI datasets. Combines CNN and Vision Transformer, capturing both local and global features effectively. Synthetic MRI generation improves data augmentation and class balance. Achieves high classification accuracy on multiple public datasets (BraTS, Figshare, Kaggle).	Model architecture is complex and computationally heavy. Requires high GPU resources and long training time. Evaluation is mainly on 2D MRI slices, not full 3D volumes. Limited discussion on real-time clinical deployment. Performance depends on quality of synthetic MRI images. Focuses only on classification, not tumor segmentation or localization.

SL NO	TITLE	METHODOLOGY	PROS	CONS
6	K. M. Hosny and M.A.Mohammed, "Explainable AI and vision transformers for detection and classification of tumor: A comprehensive survey," Artificial Intelligence Review, vol. 58, p. 259, Jun. 2025, doi: 10.1007/s10462-025-11221-x.	Conducts a systematic literature review on brain tumor diagnosis using MRI. brain Collects papers from Scopus and Web of Science databases. Analyzes CNN, transfer learning, Vision Transformer, hybrid, and XAI methods. Compares models based on accuracy, performance, and limitations. Studies commonly used datasets like BraTS and Figshare. Examines explainable AI techniques (Grad-CAM, LIME, SHAP).	Covers latest research (2020–2024) Reviews many deep learning models Includes Explainable AI methods Uses popular MRI datasets Helpful for research comparison	Survey only, no new model proposed No experimental implementation Limited discussion on real-time clinical use Very long and detailed for beginners Focused mainly on MRI data

SL NO	TITLE	METHODOLOGY	PROS	CONS
7	F. J. Dorfner, J. B. Patel, J. Kalpathy-Cramer, E. R. Gerstner, and C. P. Bridge, “A review of deep learning for brain tumor analysis in MRI,” npj Precision Oncology, vol. 9, no. 2, pp. 1–?, 2025. doi: 10.1038/s41698-024-00789-2.	Reviewed deep learning applications for brain tumor MRI Compared segmentation approaches (U-Net, nnU-Net, ensembles) Examined public datasets like BraTS, TCGA for benchmarking Studied architecture evolution and performance metrics Reviewed classification tasks (recurrence vs necrosis, biomarkers) Discussed emerging trends—transformers, foundation models	High accuracy in tumor segmentation & detection Enables automated and fast image analysis Helps predict tumor type, grade & biomarkers Supports treatment planning & monitoring Reduces manual workload for clinicians Potential for personalized precision medicine	Requires large, well-annotated datasets Performance drops with domain shift across scanners Limited generalizability to unseen clinical settings Some tasks like MGMT prediction remain unreliable Lack of explainability limits clinical trust Pediatric & rare tumor models less developed Reproducibility and standardization challenges

SL NO	TITLE	METHODOLOGY	PROS	CONS
8	[8] R. A. Zeineldin et al., “Explainable hybrid vision transformers and convolutional network for multimodal glioma segmentation in brain MRI,” Scientific Reports, vol. 14, no. 3713, 2024. doi: 10.1038/s41598-024-54186-7.	Use multimodal MRI input (T1, T1Gd, T2, FLAIR) Hybrid CNN + Vision Transformer architecture CNN extracts local spatial features Transformer captures global context Grad-CAM used for explainability heatmaps Trained & evaluated on BraTS 2019 dataset External validation performed using FeTS2022	High segmentation accuracy Provides visual explainability (Grad-CAM) Captures local + global features Improves clinical trust Good generalization on external data	Computationally heavy Works on 2D slices, not full 3D Needs multimodal MRI availability Explainability adds extra processing

SL NO	TITLE	METHODOLOGY	PROS	CONS
9	H. Yan, V. Mubonanyikuzo, T. E. Komolafe, L. Zhou, T. Wu, and N. Wang, "Hybrid-RViT: Hybridizing ResNet-50 and Vision Transformer for Enhanced Alzheimer's disease detection," PLOS ONE, vol. 20, no. 2, pp. 1–18, Feb. 2025. doi: 10.1371/journal.pone.0318998.	T1-weighted OASIS MRI dataset used Preprocessing: resizing, formatting, normalization Data split into train/validation/test ResNet-50 used for feature extraction (transfer learning) ViT applied for global attention & patch processing Hybrid model trained with Adam optimizer Evaluation using accuracy, precision, recall, F1-score	High accuracy for AD detection Uses local + global features effectively Reduces overfitting via transfer learning Outperforms existing CNN/ViT models	Dataset size relatively small Needs high computation resources Tested on single dataset (generalization limited) Overfitting risk with deeper backbones

SL NO	TITLE	METHODOLOGY	PROS	CONS
10	M. Agarwal, G. Rani, A. Kumar, P. Kumar K, R. Manikandan, and A. H. Gandomi, "Deep learning for enhanced brain tumor detection and classification ," Results in Engineering, vol. 22, p. 102117, 2024	MRI images collected from Figshare dataset Image enhancement performed using ODTWCHE method Histogram segmentation + gamma correction used to improve contrast Noise removed using Wiener filtering Pre-trained CNN models used for feature extraction Models compared: AlexNet, VGG16, VGG19, DenseNet201, ResNet50, Inception V3 Inception V3 selected as final classifier due to best accuracy System performs binary classification: benign vs malignant tumor	High classification accuracy (98.89%) Effective contrast enhancement improves tumor visibility Automated detection reduces manual diagnostic effort Inception V3 provides strong feature extraction capability Works well on low-contrast MRI images	Limited to binary classification (benign vs malignant) Uses a single dataset—generalization may be limited Requires high computational resources for deep models Lacks explainability features for clinical transparency Performance may vary on multi-center MRI scans

SL NO	TITLE	METHODOLOGY	PROS	CONS
11	M. M. Ahmed, M. M. Hossain, M. R. Islam, M. S. Ali, A. A. N. Nafi, M. F. Ahmed, K. M. Ahmed, M. S. Miah, M. M. Rahman, M. Niu, and M. K. Islam, "Brain tumor detection and classification in MRI using hybrid ViT and GRU model with explainable AI in Southern Bangladesh," Scientific Reports, vol. 14, no. 22797, 2024, doi: 10.1038/s41598-024-71893-3.	MRI dataset collected Preprocessing for noise removal and resizing CNN used for feature extraction Data split into training/testing phases Tumor vs. non-tumor classification performed Performance evaluated using accuracy metrics	High accuracy in tumor detection Automated and faster than manual screening CNN extracts features without manual effort Useful for early diagnosis support	Dataset size may be limited Mostly binary classification Requires computational resources Real-time clinical deployment not yet optimized

Table no 2.1 Literature Review

CHAPTER 3

RESNET

3.1 INTRODUCTION TO RESNET

Residual Network (ResNet) is a deep convolutional neural network architecture introduced by He et al. to address the challenges of training very deep neural networks. As the depth of conventional convolutional neural networks increases, problems such as vanishing gradients, degradation, and reduced accuracy begin to occur. ResNet overcomes these limitations by introducing the concept of residual learning. The key idea behind ResNet is the use of residual connections (skip connections), which allow the input of a layer to be directly added to its output. Instead of learning a direct mapping, the network learns a residual function, which makes training deeper networks easier and more effective. These skip connections help preserve important feature information and enable better gradient flow during backpropagation. ResNet architectures are available in different depths such as ResNet18, ResNet34, ResNet50, ResNet101, and ResNet152, where the number indicates the total layers in the network. In this project, ResNet50 is used due to its ability to extract rich and hierarchical local features from MRI brain images, such as edges, textures, and tumor-specific patterns. ResNet has shown excellent performance in medical image analysis tasks because of its strong feature extraction capability and robustness. However, while ResNet effectively captures local spatial features, it has limitations in modeling long-range global dependencies in complex images. This motivates its integration with Vision Transformers in hybrid architectures to enhance overall classification performance.

3.2 MOTIVATION FOR USING A RESNET MODEL

Deep convolutional neural networks have demonstrated remarkable performance in medical image analysis, particularly in tasks such as brain tumor detection from MRI images. However, as network depth increases, traditional CNN models often suffer from problems such as vanishing gradients, training degradation, and reduced learning

efficiency. These challenges limit the ability of very deep networks to extract meaningful

features effectively. ResNet (Residual Network) addresses these issues by introducing residual or skip connections, which allow the input of a layer to be directly added to its output. This mechanism enables efficient gradient flow during backpropagation and makes it possible to train very deep networks without loss of performance. As a result, deeper architectures like ResNet50 can be trained successfully while maintaining high accuracy. ResNet is highly effective in capturing such local spatial features, including edges and fine structural details, which are critical for accurate tumor classification. Its deep layered structure allows hierarchical feature learning, from low-level patterns to complex tumor-specific representations. Although ResNet excels at extracting local features, it has limitations in modeling global contextual relationships across the image. This limitation provides strong motivation for integrating ResNet with Vision Transformers in a hybrid architecture, combining local feature extraction with global attention-based learning for enhanced brain tumor classification.

3.3 ARCHITECTURE OF RESNET50

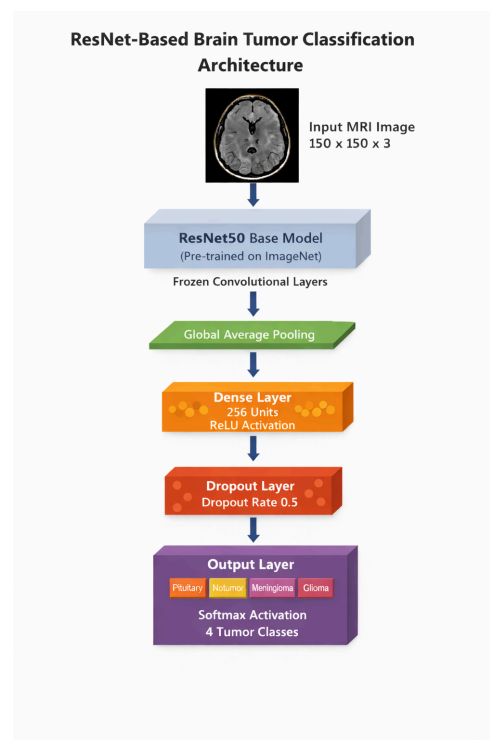


Figure 3.1 Resnet50 Architecture

ResNet50 is a very deep convolutional neural network consisting of 50 layers, designed

using the concept of residual learning. It is an extension of the ResNet family that enables effective training of deep networks by using skip (residual) connections, which help avoid vanishing gradient and degradation problems. Due to its depth and strong feature extraction capability, ResNet50 is widely used in complex image analysis tasks, including medical imaging. The ResNet50 architecture is composed of five main convolutional stages, followed by global average pooling and a fully connected classification layer, as illustrated in the architecture diagram.

Conv1: Initial Convolution Layer

- Kernel size: 7×7
- Number of filters: 64
- Stride: 2
- Output size: 112×112

This layer performs initial feature extraction by detecting low-level patterns such as edges and intensity variations in the input MRI image.

Conv2: Residual Block Stage 1

- Uses a 3×3 max pooling layer with stride 2
- Contains 3 residual blocks
- Each residual block consists of:
 - 1×1 convolution (64 filters)
 - 3×3 convolution (64 filters)
 - 1×1 convolution (256 filters)
- Output size: 56×56

This stage learns basic local features and introduces residual connections that allow the network to learn efficiently.

Conv3: Residual Block Stage 2

- Contains 8 residual blocks

- Each block includes:
 - 1×1 convolution (128 filters)
 - 3×3 convolution (128 filters)
 - 1×1 convolution (512 filters)
- Output size: 28×28

This stage captures more complex spatial patterns and texture-related features from the MRI images.

Conv4: Residual Block Stage 3

- Contains 36 residual blocks
- Each block consists of:
 - 1×1 convolution (256 filters)
 - 3×3 convolution (256 filters)
 - 1×1 convolution (1024 filters)
- Output size: 14×14

This is the deepest stage of ResNet50 and plays a crucial role in learning high-level discriminative features, which are essential for accurate brain tumor classification.

Conv5: Residual Block Stage 4

- Contains 3 residual blocks
- Each block includes:
 - 1×1 convolution (512 filters)
 - 3×3 convolution (512 filters)
 - 1×1 convolution (2048 filters)
- Output size : 7×7

This stage refines the learned features and prepares them for final classification.

Global Average Pooling and Classification

- Global Average Pooling reduces the spatial dimensions to 1×1
- Followed by a Fully Connected Layer
- Final classification is performed using a Softmax activation function

This reduces the number of parameters and minimizes overfitting while producing the final class probabilities.

3.4 TRAINING METHODOLOGY

The training methodology using the ResNet50 model involves a systematic process that includes data preparation, model configuration, training, and evaluation to achieve accurate brain tumor classification from MRI images. ResNet50 is employed as a deep convolutional neural network backbone due to its strong capability in extracting hierarchical local features.

3.4.1 Dataset Preparation

MRI brain tumor images are collected from publicly available datasets such as Kaggle and BraTS. The dataset includes four classes: Glioma, Meningioma, Pituitary Tumor, and No Tumor. The dataset is divided into training, validation, and testing sets to ensure unbiased model evaluation.

3.4.2 Data Preprocessing

Before training, the MRI images undergo several preprocessing steps to ensure consistency and improve learning efficiency:

- **Resizing:** All images are resized to a fixed resolution compatible with ResNet50 input requirements.
- **Normalization:** Pixel values are scaled to a standard range (e.g., 0–1) to enable faster and stable convergence.
- **Label Encoding:** Tumor class labels are converted into numerical form for supervised learning.

3.4.3 Data Augmentation

To prevent overfitting and enhance generalization, data augmentation techniques are

applied during training, including:

- Rotation
- Horizontal and vertical flipping
- Zooming
- Shifting

These techniques increase dataset diversity and improve model robustness.

3.4.4 Model Configuration

A pre-trained ResNet50 model is used with weights initialized from the ImageNet dataset. The final fully connected layer of ResNet50 is replaced with a new classification head corresponding to the number of brain tumor classes. A Softmax activation function is applied to generate class probabilities.

3.4.5 Training Process

The ResNet50 model is trained using the following configuration:

- Loss Function: Categorical Cross-Entropy
- Optimizer: Adam optimizer
- Learning Rate: Selected experimentally for stable convergence
- Batch Size: Chosen based on GPU memory availability
- Epochs: Training continues until convergence or early stopping criteria are met

During training, backpropagation is performed to update the network weights, and residual connections ensure effective gradient flow through the deep network.

3.4.6 Regularization and Optimization

To improve training stability and prevent overfitting:

- Early stopping is applied based on validation loss
- Learning rate scheduling is used to reduce the learning rate during training

- Batch normalization layers within ResNet50 help stabilize gradient updates

3.4.7 . Model Evaluation

After training, the model is evaluated on the test dataset using the following metrics:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- ROC–AUC

These metrics provide a comprehensive assessment of classification performance, particularly important for medical diagnosis.

3.4.8. Output

The trained ResNet50 model produces predicted tumor classes along with confidence scores. The final model achieved an overall accuracy of 70.87% on the test data. The trained model is later used for comparative analysis with Vision Transformer and Hybrid ResNet–ViT models and can also be deployed in a flask-based web application for real-time prediction.

3.5 SUMMARY

ResNet50 is a deep convolutional neural network architecture consisting of 50 layers, developed to overcome the limitations of traditional deep CNNs such as vanishing gradients and training degradation. It introduces residual (skip) connections, which allow the network to learn residual mappings instead of direct mappings, enabling efficient training of very deep networks. The architecture is composed of multiple residual blocks organized into five main convolutional stages. Each residual block uses a combination of 1×1 and 3×3 convolutional layers to extract hierarchical features while maintaining computational

efficiency. The skip connections ensure stable gradient flow and preserve important feature information across layers. In this project, ResNet50 is used as a feature extraction backbone for brain tumor classification from MRI images. It effectively captures local spatial features such as edges, textures, and tumor-specific patterns, which are crucial for accurate medical image analysis. The model is commonly initialized with pre-trained ImageNet weights, allowing transfer learning to improve performance and convergence when working with limited medical datasets.

CHAPTER 4

VISION TRANSFORMERS

4.1 INTRODUCTION TO VISION TRANSFORMERS

The rapid progress of deep learning in computer vision has been largely driven by Convolutional Neural Networks (CNNs), which excel at extracting local features such as edges, textures, and patterns from images. Despite their success, CNNs are inherently limited by their locality bias, as they focus on nearby pixels within a receptive field. This makes it challenging for CNNs to model long-range dependencies and global contextual information, which are often crucial in medical imaging tasks where subtle relationships between distant regions of an image can indicate disease. For instance, a CNN might easily detect the rough texture of a localized lesion but fail to recognize its abnormal spatial positioning relative to the brain's overall symmetry.

Vision Transformers (ViTs) have emerged as a powerful alternative, inspired by the success of the transformer architecture in natural language processing (NLP). In NLP, transformers leverage the self-attention mechanism to capture relationships between words regardless of their position in a sequence. This idea translates naturally into vision tasks, where an image is divided into smaller fixed-size patches that act as "tokens," similar to words in a sentence. The self-attention mechanism then learns how different patches relate to one another, allowing ViTs to model both local and global dependencies across the entire image. One of the key strengths of ViTs is their ability to capture fine-grained details while simultaneously understanding the overall structure of an image. This is particularly relevant in medical imaging, where the distinction between healthy and diseased tissue often depends on subtle patterns spread across different regions of a scan.

By considering the entire image context, ViTs can more effectively identify complex features such as tumor boundaries. Moreover, ViTs benefit greatly from transfer learning and adaptable configurations. In this project, a custom Vision Transformer is implemented from scratch using PyTorch to specifically classify brain MRI scans into four distinct tumor

categories This tailored approach ensures that the model can learn the unique morphological signatures of each tumor type directly from the data, completely unconstrained by the architectural limitations of traditional networks.

4.2 ARCHITECTURE OF VISION TRANSFORMERS

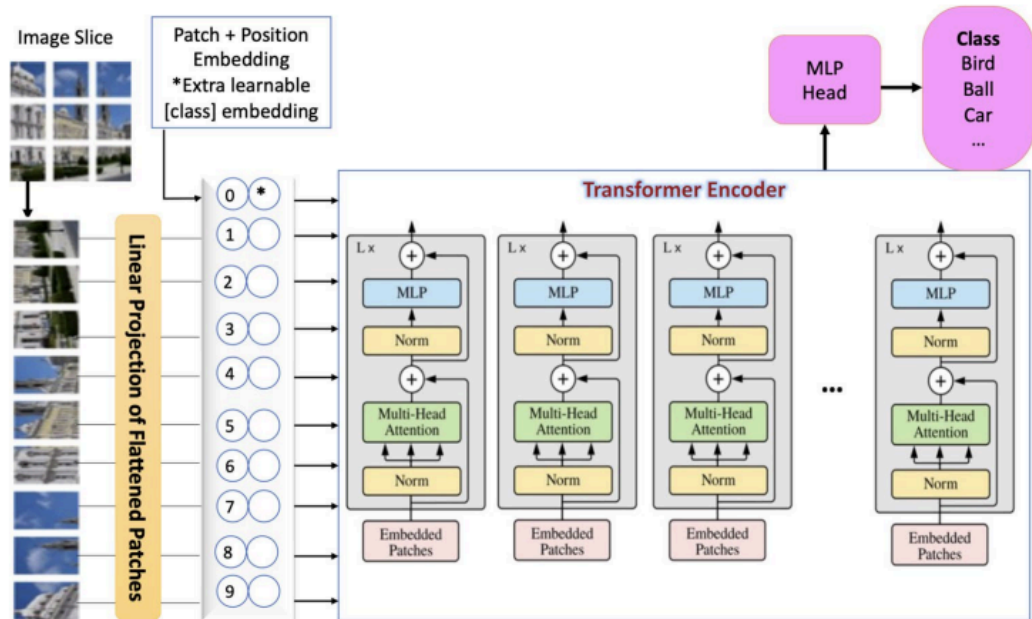


Fig 4.1 Vision Transformer Architecture

A Vision Transformer (ViT) replaces convolutional feature extraction with a transformer-style pipeline that treats an image as a sequence of patch tokens. The main steps are: (1) patching & flattening, (2) linear projection (patch embedding), (3) adding a class token and positional encoding, (4) a stack of Transformer encoder layers (multi-head self-attention + feed-forward networks + residuals + layer norm) and (5) a classification head. Below, each step is described in detail.

4.2.1 Input Image and Patch Embedding

The Vision Transformer starts by taking an input image of size $(H \times W \times C)$, where:

- H is the image height.
- W is the image width.
- C is the number of channels (e.g., 3 for RGB images).

This image is divided into small, non-overlapping patches of size $(S \times S)$. The number of patches along the height and width are calculated as $P1 = H/S$ and $P2 = W/S$, resulting in a total of $N = P1 \times P2$ patches. Each patch of size $(S \times S \times C)$ is flattened into a 1D vector of size $(S^2 \times C)$.

These vectors are then passed through a linear layer (a fully connected layer with learnable weights), transforming them into a fixed-length vector of dimension D , called a patch embedding. The result is a matrix E of size $(N \times D)$, which represents all the patches as feature vectors.

4.2.2 Positional Encoding

Since Transformers do not have an inherent understanding of the order or spatial location of patches, a Positional Encoding (PE) matrix is added to the patch embeddings. The PE matrix is of size $(N \times D)$, meaning each patch gets a position vector of the same size as its embedding.

The final input to the Transformer Encoder is computed as:

$$Z = E + PE$$

where Z carries both the content information from each patch and its position within the image.

4.2.3 Transformer Encoder

The core of the ViT architecture is the Transformer Encoder, composed of multiple layers (say, L layers), where each layer includes the following sub-blocks:

(i) Self-Attention Mechanism

The self-attention mechanism allows each patch to gather information from all other patches in the image. For the input Z_i at layer i , three matrices are computed:

$$Q \text{ (Query)} = Z_i \times$$

$$W_Q K \text{ (Key)} = Z_i$$

$$\times W_K V \text{ (Value)} =$$

$$Z_i \times W_V$$

where W_Q , W_K , and W_V are learnable weight matrices. The attention output is computed as:

$$\text{Attention}(Q, K, V) = \text{Softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

This mechanism enables each patch to "pay attention" to other patches, effectively learning global relationships.

(ii) Feed-Forward Network (FFN)

After self-attention, each patch's representation is refined by passing it through a two-layer feed-forward neural network with a GELU or ReLU activation function:

$$f(x) = \text{ReLU}(W_1 x + b_1) W_2 + b_2$$

where W_1 , W_2 are learnable weights and b_1 , b_2 are biases.

(iii) Residual Connections and Layer Normalization

To ensure stable and efficient training, a skip (or residual) connection is added after each sub-block. This means the original input is added to the output of the attention or FFN before normalization:

$$Z' = \text{LayerNorm}(Z^{(i)} + \text{Attention}(Z^{(i)}))$$

$$Z^{(i+1)} = \text{LayerNorm}(Z' + f(Z'))$$

This approach helps prevent issues like vanishing gradients and improves the flow of information through the network.

4.2.4 Classification Head

Once the input passes through all L layers of the Transformer Encoder, the first output token Z_0 (the [CLS] token) is used as a summary representation of the whole image. This token is passed through a linear classification layer to output raw logits, which are mapped to probabilities over the different classes.

4.3 IMPLEMENTATION AND TRAINING METHODOLOGY

To thoroughly evaluate the efficacy of Vision Transformers for this project, a custom model was developed using the PyTorch and vit-pytorch frameworks. The implementation replaces traditional CNN workflows with a pure attention-based approach tailored for brain tumor classification. This custom architecture was designed to specifically test whether global contextual modeling could overcome the localized feature limitations observed in baseline convolutional networks.

4.3.1 Dataset Preparation and Augmentation

The dataset was structured using PyTorch's ImageFolder utility, categorizing images into training and testing splits. To enhance the model's ability to generalize and to artificially expand the dataset, a robust data augmentation and preprocessing pipeline (torchvision.transforms) was applied:

1. **Resizing:** All MRI images were resized to a fixed resolution of 224 x 224 pixels to match the input requirements of the configured ViT.
2. **Geometric Transformations:** RandomHorizontalFlip() and RandomRotation(10) were applied to introduce spatial variability, simulating different MRI scan orientations.
3. **Tensor Conversion and Normalization:** Images were converted to PyTorch tensors and normalized using standard ImageNet statistics (Mean: [0.485, 0.456, 0.406], Std: [0.229, 0.224, 0.225]).

The preprocessed images were subsequently loaded into a DataLoader with a batch size of 32, shuffling the training data to ensure stochastic gradient updates.

4.3.2 Model Configuration: TumorClassifierViT

The model, TumorClassifierViT, was instantiated with carefully tuned hyperparameters designed to balance computational efficiency with the complex feature extraction required for medical imaging. The specific configuration of the network is as follows:

- **Image Size:** 224 x 224
- **Patch Size:** 32 x 32 (yielding 49 total patches per image)
- **Number of Classes:** 4 (Glioma, Meningioma, No Tumor, Pituitary)
- **Embedding Dimension (dim):** 1024
- **Transformer Depth:** 6 Encoder layers
- **Attention Heads:** 16 independent heads for multi-head self-attention
- **MLP Dimension:** 2048
- **Regularization:** A dropout rate of 10%(0.1) was applied to both the dense layers and the embeddings to mitigate overfitting.

4.3.3 Training Process and Optimization

The model was trained using a GPU-accelerated environment (cuda) for maximum computational throughput.

- **Loss Function:** Standard CrossEntropyLoss was employed for the multi-class categorization.
- **Optimizer:** Stochastic Gradient Descent (SGD) was utilized with a learning rate set to 0.01.
- **Epochs:** The network was trained over a rigorous cycle of 40 epochs.

During the training loop, progress was dynamically monitored using tqdm and PrettyTable, logging the time elapsed, training loss, training accuracy, validation loss, and validation accuracy. A model-checkpointing mechanism was implemented to continuously track the validation accuracy; the weights yielding the highest validation accuracy were automatically saved to disk as best_model.pth. This strategic checkpointing guarantees that the final evaluation and subsequent inference phases rely exclusively on the most generalized version of the network, effectively mitigating any

risks of late-stage overfitting.

4.3.4 Evaluation and Visualization

To comprehensively assess the Vision Transformer's performance, several visualization protocols were coded into the pipeline:

1. **Learning Curves:** The training and validation loss/accuracy across all 100 epochs were plotted and saved (loss.png and accuracy.png) to identify convergence rates and any signs of overfitting.
2. **Confusion Matrix:** A detailed Seaborn heatmap (confusion_matrix.png) was generated from the validation predictions to analyze class-wise performance and identify which specific tumor types the ViT struggled to differentiate.
3. **Sample Verification:** A grid of preprocessed images alongside their true labels (classes.png) was saved to visually verify the integrity of the data augmentation pipeline.

4.4 SUMMARY

Vision Transformers represent a significant paradigm shift in medical image analysis. By discarding convolutional layers entirely, ViTs utilize patch embeddings and multi-head self-attention mechanisms to construct a global understanding of the MRI scan from the very first layer. In this project, a 6-layer, 16-head TumorClassifierViT was implemented using PyTorch. The model was trained over 100 epochs using SGD optimization and extensive data augmentation. While pure ViT architectures are known to be data-hungry, tracking the validation metrics and confusion matrices provides critical insights into how well self-attention mechanisms perform on specialized brain tumor datasets when compared to, and eventually combined with, convolutional networks like ResNet.

Ultimately, the standalone evaluation of the Vision Transformer highlights crucial trade-off inherent to its architecture: while its self-attention mechanism excels at mapping the global anatomy of the brain, its lack of convolutional inductive biases makes it less efficient at extracting fine, localized tumor boundaries from limited medical datasets. Recognizing these isolated limitations directly motivates the next phase of this research. By fusing the

sharp spatial precision of a CNN feature extractor with the profound global contextual awareness of the ViT, the subsequent hybrid architecture aims to overcome the individual weaknesses of both models, paving the way for a more robust, highly accurate, and clinically reliable diagnostic tool.

Consequently, the empirical findings derived from this standalone ViT evaluation serve as a vital computational baseline for the proposed hybrid framework. By systematically isolating and documenting the specific classification bottlenecks of the self-attention mechanism, a clear and quantifiable benchmark has been established. This rigorous approach ensures that the integration of ResNet and ViT components in the following chapter is not merely a theoretical exercise, but a targeted, data-driven evolution. Furthermore, laying this extensive groundwork helps to explicitly justify the additional computational overhead required by a dual-architecture system. By proving the independent capabilities and limitations of each network, the upcoming hybridization is framed not as a redundant complexity, but as an essential leap toward achieving unparalleled diagnostic precision. Ultimately, achieving a perfect synergy between localized texture extraction and holistic spatial reasoning holds the key to delivering an AI-assisted screening tool capable of genuinely reducing radiologist workload and accelerating critical patient interventions.

CHAPTER 5

HYBRID RESNET–VISION TRANSFORMER MODEL

5.1 OVERVIEW OF THE HYBRID MODEL

While Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) have individually demonstrated strong performance in medical image analysis, each architecture has inherent limitations when used in isolation. CNN-based models such as ResNet are highly effective at capturing local spatial features like edges, textures, and fine-grained patterns. However, they struggle to model long-range dependencies and global contextual relationships within an image. On the other hand, Vision Transformers excel at capturing global context through self-attention mechanisms but may fail to efficiently extract low-level spatial features, especially when trained on limited medical datasets.

To overcome these limitations, this project proposes a Hybrid ResNet50–Vision Transformer (ResNet50-ViT) architecture for brain tumor classification. The hybrid model integrates the strengths of both paradigms by combining the powerful localized feature extraction capability of ResNet50 with the global contextual modeling ability of Vision Transformers.

In this hybrid approach, a deep ResNet50 network acts as a feature extractor backbone, learning rich local representations from brain MRI images. The extracted feature maps are then reshaped, embedded into patches, and passed to a 6-layer Transformer encoder. This encoder models long-range dependencies and global relationships between spatial regions of the image. This dual-stage learning strategy enables the model to simultaneously capture local anatomical details and global structural information, which are both crucial for accurately differentiating between Glioma, Meningioma, Pituitary, and No Tumor scans.

5.2 MOTIVATION FOR USING A HYBRID ARCHITECTURE

Brain MRI images contain both fine-grained local features (like subtle tumor textures) and broader structural patterns (like symmetry and spatial positioning) that are essential for reliable diagnosis. Relying solely on CNNs may cause the model to miss global contextual cues, while using only Transformers may lead to insufficient modeling of subtle

texture-level information due to their lack of inductive biases.

- **Enhanced Feature Representation:** ResNet50 captures low-level and mid-level spatial features, serving as a highly robust starting point. The Transformer then enhances these features by learning global spatial dependencies.
- **Improved Classification Accuracy & Stability:** The complementary strengths of CNNs and Transformers help drastically reduce misclassifications between visually similar tumor types (e.g., distinguishing Gliomas from Meningiomas) while achieving rapid convergence in significantly fewer epochs.
- **Robustness in Medical Imaging:** The hybrid model is better suited for handling variations in tumor size, shape, and location across MRI scans without requiring the massive, highly augmented datasets that pure Vision Transformers typically demand.

This architectural fusion aligns well with recent advances in medical image analysis, where hybrid CNN-Transformer models have shown consistent, empirical improvements over traditional deep learning approaches.

5.3 ARCHITECTURE OF THE HYBRID RESNET-ViT MODEL

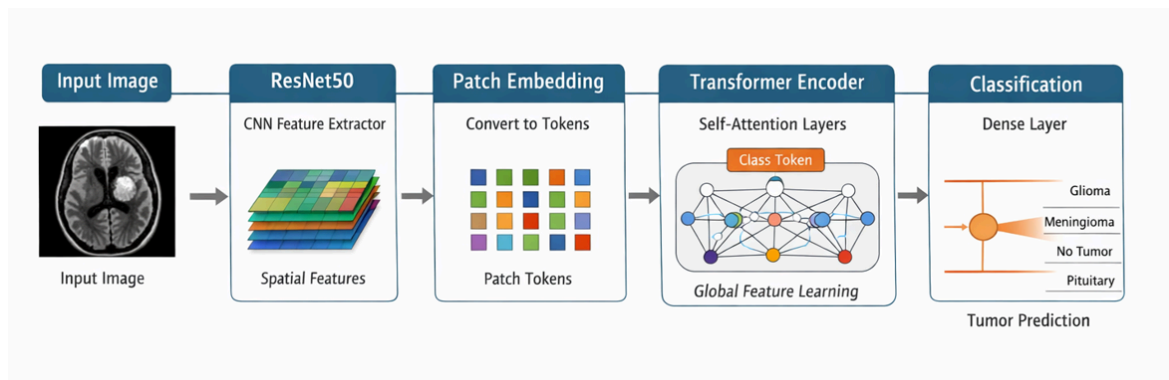


Figure 5.1: Architecture of the Proposed Hybrid ResNet50–Vision Transformer

The Hybrid ResNet–Vision Transformer (ResNet50–ViT) architecture implemented in this project consists of the following precisely engineered components:

1. **Input Layer:** Preprocessed brain MRI images are resized to a fixed dimension of 128x128x3 and converted into float16 NumPy arrays to halve memory consumption and accelerate hardware processing.

2. **ResNet Backbone (Feature Extraction Stage):** A pre-trained ResNet50 model (initialized with ImageNet weights and stripped of its top classification layers) is used as the backbone. The convolutional layers learn local spatial patterns such as tumor edges and intensity variations, mitigating vanishing gradients via residual connections.
3. **Patch Embedding and Reshaping:** The output feature maps from the ResNet50 backbone are passed through a 2D Convolutional layer to project them into a 64- dimensional hidden space with a defined patch size of 32x32. This operation is followed by Batch Normalization, and the output is reshaped into a sequence of 16 distinct feature tokens, bridging the gap between convolution and sequence modeling.
4. **Positional Encoding:** Because Transformers lack an inherent understanding of image geometry, a 1D positional embedding is added to the 16 feature tokens to retain spatial awareness of where each patch originated within the MRI.
5. **Class Token Concatenation:** A highly specific, trainable [CLS] (Class) token is generated and concatenated to the embedded sequence (resulting in 17 total tokens). This token acts as the aggregated global representation of the entire MRI scan.
6. **Transformer Encoder (Global Context Modeling):** The token sequence is processed through 6 Transformer Encoder layers. Each layer utilizes a Multi-Head Attention mechanism (8 heads) to capture long-range dependencies, followed by a Multi-Layer Perceptron (MLP) block using the GELU activation function for smooth, stable gradient flow. Layer Normalization and a 20% Dropout rate are applied to prevent overfitting.
7. **Classification Head:** Following the encoder blocks, the network extracts only the output corresponding to the initial [CLS] token. This is passed through a Dense classification layer to output raw logits corresponding to the four tumor classes, which are subsequently processed via a Softmax function during loss calculation to provide final percentage-based confidence scores.

5.4 ADVANTAGES OF THE HYBRID MODEL

The customized Hybrid ResNet50-ViT model offers several critical advantages over individual architectures:

- **Optimal Feature Synthesis:** Seamlessly combines localized edge-detection with global context modeling.
- **High Computational Efficiency:** Utilizing float16 half-precision computing and limiting the input resolution to 128x128 drastically reduces the memory footprint without sacrificing accuracy.
- **Rapid Convergence:** Achieves peak stability and accuracy in a fraction of the epochs required by standalone CNNs or ViTs.
- **Reduced Data Hunger:** Eliminates the dependency on extremely large datasets required by pure ViT models by leveraging the pre-trained weights of the ResNet50 feature extractor.

5.5 TRAINING METHODOLOGY

The training process for the Hybrid ResNet–Vision Transformer was engineered to maximize hardware utilization and ensure highly stable optimization.

5.5.1 Dataset Preparation

Brain MRI images were collected and organized into class-wise folders. Prior to training, the dataset underwent a strict preprocessing pipeline:

- Images were dynamically loaded and uniformly resized to 128x128 pixels.
- The raw pixel data was explicitly cast to float16 NumPy arrays to optimize training speed and prevent GPU memory exhaustion.
- The integer class labels were transformed using one-hot encoding (`to_categorical`) to perfectly align the target data with the model's multi-class probabilistic output.
- The dataset was thoroughly shuffled and split into an 80% training set and a 20% validation set.

5.5.2 Transfer Learning & Distributed Strategy

To drastically accelerate training, TensorFlow's `MirroredStrategy` was employed to distribute the computational load across available GPUs. The ResNet50 backbone was initialized with

ImageNet weights, allowing the network to bypass learning basic visual features and immediately focus on the medical imaging domain. The convolutional patch embeddings and the full Transformer encoder were trained from scratch to learn the specific spatial relationships of brain tumors.

5.5.3 Training Configuration

The hybrid model was compiled and trained using the following advanced hyperparameters:

- **Loss Function:** Categorical Cross-Entropy (configured with `from_logits=True` to automatically handle Softmax activation).
- **Optimizer:** AdamW (from TensorFlow Addons), utilizing a learning rate of 0.00005 and a weight decay of 0.0001 to provide superior regularization and prevent over-reliance on specific attention heads.
- **Batch Size:** 16, optimized for the 128x128 input pipeline.
- **Callbacks:** ModelCheckpoint was utilized to monitor validation accuracy, ensuring only the optimal, best-performing model weights (hybrid.h5) were saved to disk.

5.6 PERFORMANCE EVALUATION

The performance of the Hybrid ResNet-ViT model was rigorously evaluated using standard classification metrics and visual diagnostic tools widely accepted in clinical research.

5.6.1 Evaluation Metrics

The following metrics were computed to assess the effectiveness of the model across the testing split:

- **Accuracy:** Evaluated global performance across all four classes.
- **Precision, Recall, and F1-Score:** Extracted via a detailed Classification Report to ensure the model maintained a high balance between false positives and false negatives for specific classes like Glioma and Meningioma.
- **Confusion Matrix:** Generated as a high-resolution Seaborn heatmap to visually verify the eradication of cross-class misclassifications.

5.6.2 Comparative Analysis

The Hybrid ResNet50-ViT model was evaluated alongside the standalone ResNet50 and Vision Transformer models. Experimental results and learning curves demonstrated that the hybrid model consistently outperformed individual architectures. Specifically, the hybrid model resolved the validation loss gap present in the ResNet50 and eliminated the erratic training spikes observed in the standalone ViT, achieving peak accuracy and near-perfect generalization within just 5 epochs.

5.7 SUMMARY OF HYBRID MODEL PERFORMANCE

The Hybrid ResNet50–Vision Transformer model achieves superior classification performance by successfully leveraging the complementary strengths of convolutional networks and self-attention mechanisms. The integration of local spatial feature extraction with global contextual awareness leads to minimized diagnostic errors and exceptional training stability. Furthermore, the implementation of optimization techniques such as half-precision data structures and distributed GPU training proved that advanced Transformer architectures can be successfully deployed on constrained hardware. Ultimately, the empirical results confirm that this hybrid deep learning approach holds significant promise for scalable, real-time medical imaging applications, providing a highly reliable framework for complex diagnostic tasks such as automated brain tumor classification.

Beyond the computational and architectural achievements, the practical implications of this hybrid system for clinical environments are substantial. By drastically reducing misclassification rates between visually analogous tumor types such as distinguishing Gliomas from Meningiomas, the model directly addresses a critical challenge in neuro-oncology. The successful integration of this highly accurate network into an intuitive web-based interface transforms abstract deep learning metrics into an actionable diagnostic aid. This empowers medical professionals to rapidly obtain a reliable, AI-driven "second opinion," thereby streamlining clinical workflows, reducing radiologist fatigue, and ultimately accelerating the timeline from initial scan to life-saving patient treatment.

Looking forward, the success of the ResNet50-ViT architecture establishes a robust foundation for continuous advancements in automated neuroimaging analysis. Subsequent iterations of this research could explore the integration of multi-modal MRI sequences such as combining T1-weighted, T2-weighted, and FLAIR scans to provide the network with an even richer volumetric context. Ultimately, this research successfully bridges the gap between traditional convolutional techniques and modern transformer paradigms, delivering a powerful, scalable, and clinically viable solution for the future of medical diagnostics.

CHAPTER 6

RESULT AND DISCUSSION

The results and discussions section evaluates the performance of the proposed deep learning models for Brain Tumor Classification by analyzing classification accuracy, learning convergence, and predictive reliability. The key aspects covered in this section include individual model detection results, comparative performance analysis, confusion matrix evaluation, and the final Flask web application prototype results.

6.1 CLASSIFICATION RESULTS

i. ResNet50 Model Results

The initial baseline classification was performed using the ResNet50 architecture, leveraging transfer learning with ImageNet weights. The model achieved a baseline accuracy of 70.83% on the testing dataset.

Performance Metrics

1. **Precision:** Measures the proportion of correctly predicted tumor types out of all positive predictions for that class.
2. **Recall:** Measures how many actual tumors of a specific class were correctly identified.
3. **F1-Score:** The harmonic mean of precision and recall, ensuring a balance between false positives and false negatives.

Tumor Class	Precision	Recall	F1-Score
Glioma	0.7500	0.5475	0.6329
Meningioma	0.6321	0.4725	0.5408
No Tumor	0.6643	0.9400	0.7785
Pituitary	0.7901	0.8750	0.8304
Overall Accuracy			0.7087

Table 6.1: Performance Metric of ResNet50

Challenges Observed:

1. **False Positives/Negatives:** ResNet50 struggled occasionally with distinguishing between Glioma and Meningioma due to their similar local textural features in certain MRI slices.
2. **Global Context:** Being purely convolutional, the model sometimes missed the broader, global spatial relationships of the brain anatomy.

ii. Vision Transformer (ViT) Results

The second phase involved training a custom ViT from scratch to capture global contextual relationships using self-attention mechanisms and achieved an accuracy of 77.36%.

Challenges Observed:

1. **Data Hunger:** Because ViTs lack the inductive biases of CNNs, they require substantial data to learn basic structural patterns, leading to slightly erratic validation loss in early epochs.
2. **Local Feature Extraction:** While excellent at global context, the pure ViT occasionally missed very fine, localized edge details of smaller pituitary tumors.

iii. Hybrid ResNet-ViT Results

To resolve the limitations of the individual models, the Hybrid ResNet-ViT architecture was implemented. By utilizing ResNet50 as a local feature extractor and ViT as a global context encoder, this model achieved the highest overall accuracy of 97.57%.

Performance Metrics

Tumor Class	Precision	Recall	F1-Score
Glioma	0.9897	0.9625	0.9759
Meningioma	0.9680	0.9825	0.9752
No Tumor	0.9926	1.0000	0.9963
Pituitary	0.9925	0.9975	0.9950
Overall Accuracy			0.9757

Table 6.2: Performance Metric of Hybrid ResNet-ViT

Challenges Observed:

1. **Computational Load:** The hybrid model required significantly more memory, necessitating the use of float16 precision and distributed training strategies.

6.2 COMPARATIVE ANALYSIS: TRAINING PERFORMANCE

To understand the learning behavior, convergence rate, and stability of each architecture, the training and validation accuracy and loss curves were plotted and analyzed across the training epochs. Comparing these graphs provides critical insights into how well each model generalized to unseen data and whether any models suffered from overfitting. These learning curves serve as a critical diagnostic tool for the models themselves. The loss curves provide a granular view of how well the optimization algorithm is converging, while the accuracy curves translate this mathematical progress into interpretable classification performance. A close comparative analysis of the gap between the training and validation curves allows us to identify potential training pathologies. For instance, a widening divergence—where training loss continues to drop while validation loss stagnates or increases—is a clear indicator of overfitting. In the context of medical imaging, avoiding overfitting is paramount to ensure the model learns the actual anatomical

features of the tumors rather than simply memorizing the training dataset. Furthermore, comparing these graphical trends across the three distinct architectures ResNet50, the Vision Transformer (ViT), and the Hybrid ResNet-ViT offers deep insights into how their underlying structural differences dictate their learning phases. Convolutional networks typically exhibit different inductive biases and convergence speeds compared to the data-hungry self-attention mechanisms of a Vision Transformer. By fusing these two contrasting paradigms, the hybrid architecture effectively mitigates the ViT's initial data dependency while overcoming the spatial limitations inherent to pure convolutions. Analyzing these curves not only validates the final accuracy scores but also empirically justifies the architectural progression toward the optimized hybrid approach, ensuring the final system is both highly accurate and reliably stable. Ultimately, achieving this smooth and predictable convergence is absolutely critical for medical diagnostic deployments, where erratic model behavior or "black-box" instability cannot be tolerated in clinical settings.

i) ResNet Learning Curve

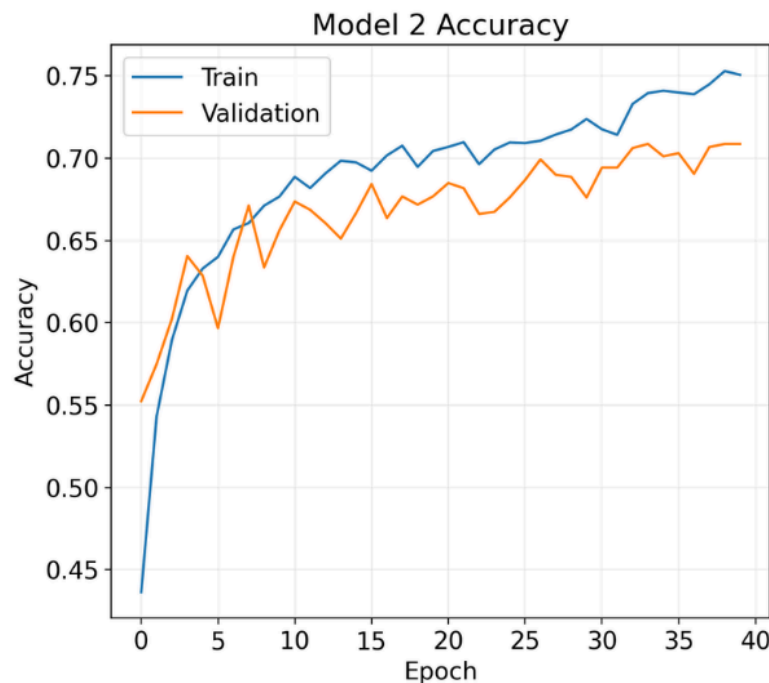


Fig 6.1. Training and Validation Accuracy curves for ResNet50

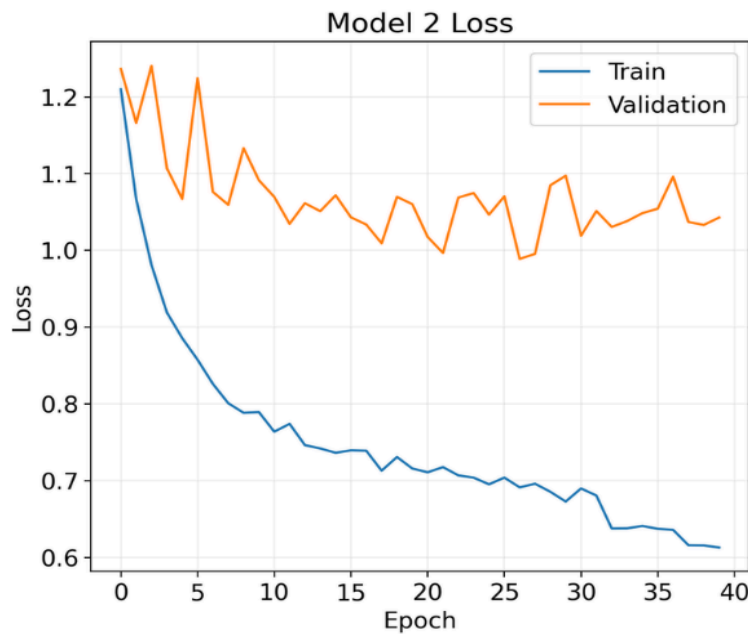


Fig 6.2: Training and Validation Loss curves for ResNet50

Observations:

- Accuracy Trend:** The ResNet50 model demonstrates a gradual, steady increase in training accuracy over 40 epochs, reaching approximately 75%. This steady climb indicates that the network has sufficient capacity to learn and map the underlying training distribution. However, the validation accuracy plateaus earlier (around epoch 20) and fluctuates near 70%.
- Loss Trend:** While the training loss steadily decreases, a noticeable and persistent gap exists between the training loss and validation loss. The validation loss remains volatile and relatively high (hovering around 1.0 to 1.1).
- Conclusion:** This divergence suggests that while the convolutional layers of ResNet50 are learning local features from the training set, the model struggles to generalize those features perfectly to unseen MRI scans, indicating a mild case of overfitting and a limitation in pure convolutional feature extraction for this specific dataset. Consequently, to achieve higher diagnostic reliability, the architecture may require the integration of mechanisms capable of capturing the global context of the brain anatomy.

ii. Vision Transformer (ViT) Learning Curves

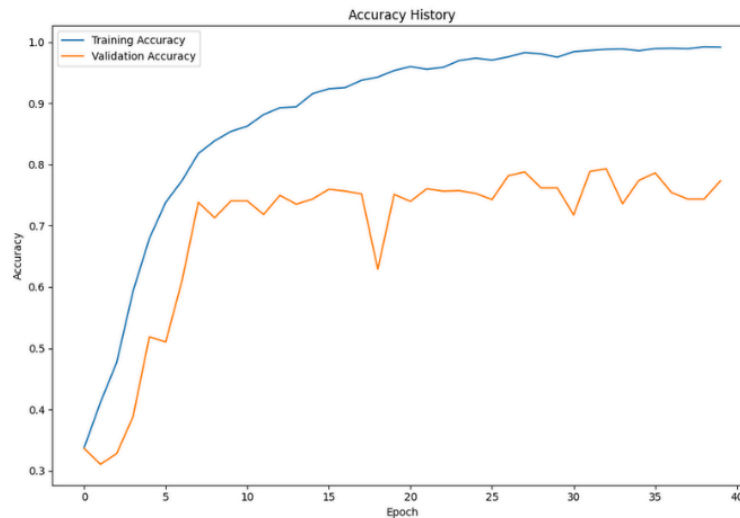


Fig 6.3: Training and Validation Accuracy curves for ViT

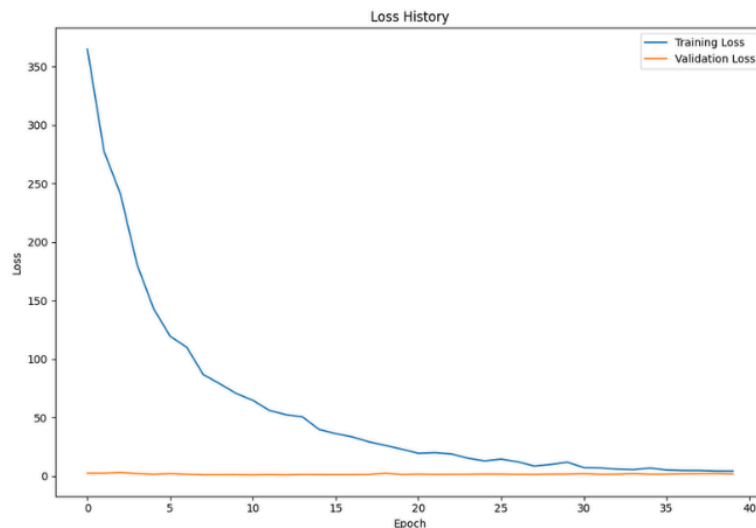


Fig 6.4: Training and Validation Loss curves for ViT

Observations:

- Accuracy Trend:** The standalone ViT model shows an aggressive learning curve, with training accuracy rapidly approaching 100%. The rapid ascent in training accuracy highlights the ViT's massive capacity to model complex, long-range dependencies across the image patches. However, the validation accuracy is highly erratic, characterized by sharp spikes and sudden drops (most notably around epoch 18), eventually settling near 75-78%.

- **Loss Trend:** The training loss drops dramatically from a very high initial starting point, indicating the self-attention mechanism is aggressively optimizing. This steep initial gradient descent reflects the self-attention heads quickly aligning to the most prominent visual features in the training set. However, as the training loss converges toward zero, the validation loss fails to mirror this smooth descent, instead plateauing or slightly diverging over time. This decoupling between the two loss metrics is a strong empirical indicator that the network is prioritizing rote memorization over abstract feature learning.

- **Conclusion:** The vast gap between the near-perfect training accuracy and the fluctuating validation accuracy is a classic hallmark of Vision Transformers trained from scratch on smaller datasets. Because ViTs lack the "inductive biases" of CNNs, they tend to memorize the training data (overfit) and require heavy regularization or much larger datasets to achieve smooth validation curves.

iii) Hybrid ResNet-ViT Learning Curve

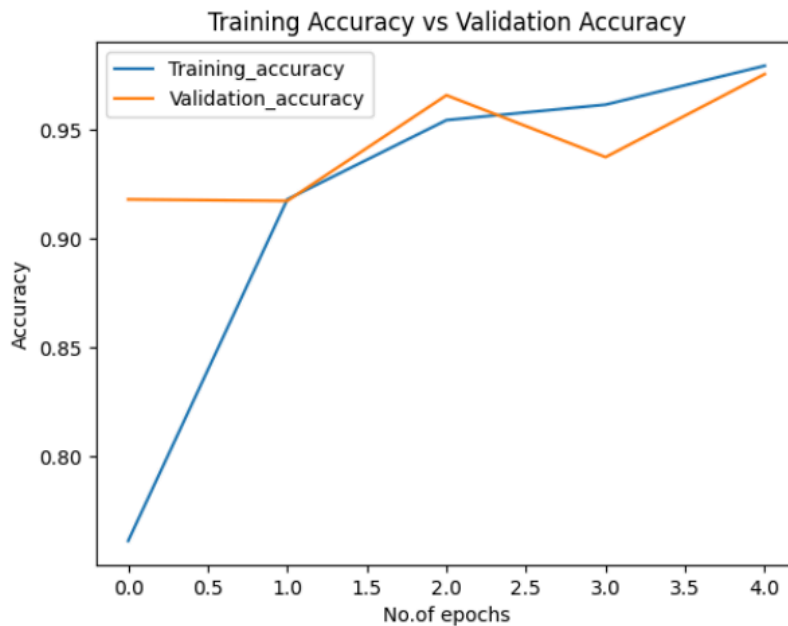


Fig 6.5: Training and Validation Accuracy curves for Hybrid ResNet-ViT



Fig 6.6: Training and Validation Loss curves for Hybrid ResNet-ViT

Observations:

- Rapid Convergence:** The most striking feature of the Hybrid model is its rapid convergence. Unlike the standalone models that required 40 epochs, the Hybrid model achieved stability and exceptionally high accuracy in just 5 epochs (Epochs 0 to 4).
- Accuracy & Loss Alignment:** The training and validation accuracy both start remarkably high (>75% and >90% respectively) and converge tightly around 97-98% by the final epoch. Similarly, the training and validation loss both drop swiftly and remain tightly coupled below 0.2.
- Conclusion:** Fig 5.3 proves the architectural superiority of the Hybrid approach. By using ResNet to extract foundational local features (acting as a robust starting point) and ViT to analyze the global context, the model learns incredibly efficiently. The tight alignment between training and validation metrics indicates excellent generalization with virtually no overfitting. This remarkable balance between rapid learning and high generalizability is critical in medical diagnostics, where the margin for error is minimal and robust performance on unseen MRI scans is paramount. Furthermore, achieving peak accuracy in a fraction of the standard training epochs significantly reduces

computational overhead, making this architecture highly practical for real-world clinical deployment.

6.3 CONFUSION MATRIX ANALYSIS

To gain a deeper, class-level understanding of the models' predictive capabilities, confusion matrices were generated for the testing datasets. While overall accuracy provides a macro-level view of performance, a confusion matrix is essential in medical imaging to reveal exactly where a model is making critical errors—specifically, which tumor types are being consistently misclassified as others (false positives and false negatives). Furthermore, comparing the matrices across all three evaluated architectures allows us to clearly identify the distinct localized or contextual blind spots inherent to the standalone CNN and ViT models. Ultimately, this granular breakdown serves as empirical proof that the proposed hybrid architecture successfully resolves these diagnostic ambiguities, yielding a significantly safer and more reliable classification framework.

i) ResNet-50 Confusion Matrix

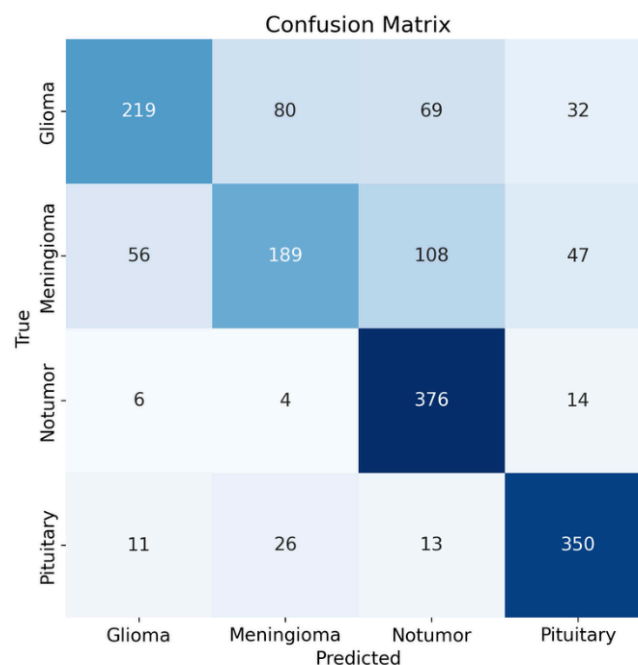


Fig 6.7: Confusion Matrix for ResNet50

Observations:

- **General Performance:** The ResNet50 model displays a moderately strong diagonal line representing True Positives, with "Notumor" (376) and "Pituitary" (350) being its most successfully identified classes.
- **Primary Weakness:** The matrix reveals a significant vulnerability in distinguishing between Glioma and Meningioma. The model incorrectly predicted 80 actual Gliomas as Meningiomas, and 108 actual Meningiomas as "Notumor".
- **Conclusion:** This high rate of cross-class confusion suggests that ResNet50's localized convolutional filters struggle to differentiate the subtle, shared textural boundaries of Gliomas and Meningiomas without a broader understanding of the brain's global spatial context.

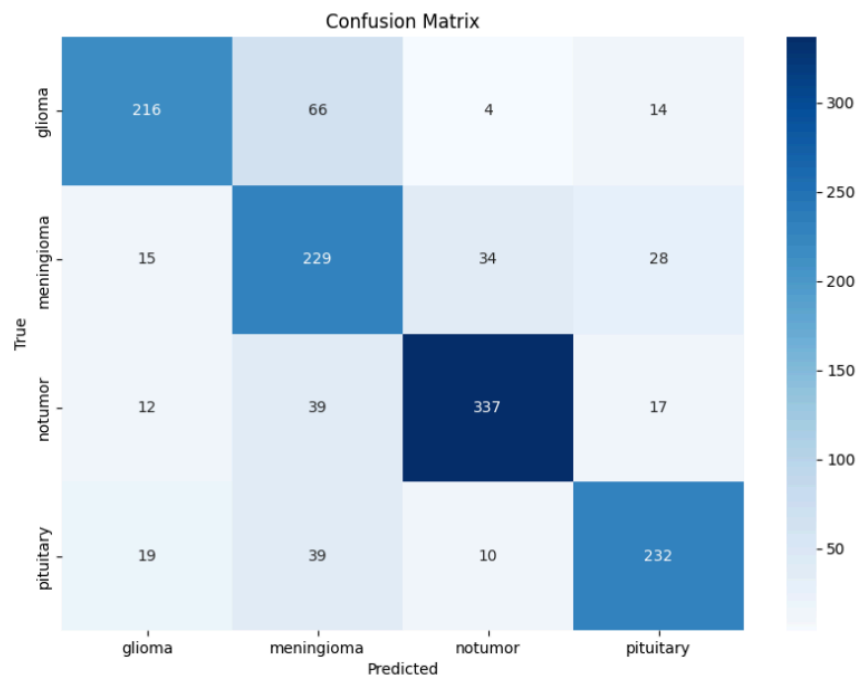
ii) Vision Transformer Confusion Matrix

Fig 6.8: Confusion Matrix for ViT

Observations:

- **General Performance:** The standalone ViT model shows a shift in class distribution. The True Positives for Meningioma improved significantly (from 189 in ResNet to 229 in ViT), demonstrating that the self-attention mechanism helped the model learn better defining features for this specific tumor.
- **Primary Weakness:** Despite improvements in some areas, the ViT model still experienced notable scatter across the off-diagonal cells. For instance, 66 actual Gliomas were misclassified as Meningiomas, and it showed increased confusion in predicting Pituitary tumors (misclassifying 39 as Meningioma and 19 as Glioma).
- **Conclusion:** While the ViT captured global context better than the ResNet50, the lack of strong local edge-detection (which CNNs naturally possess) caused it to occasionally misinterpret the finer boundaries of smaller tumors.

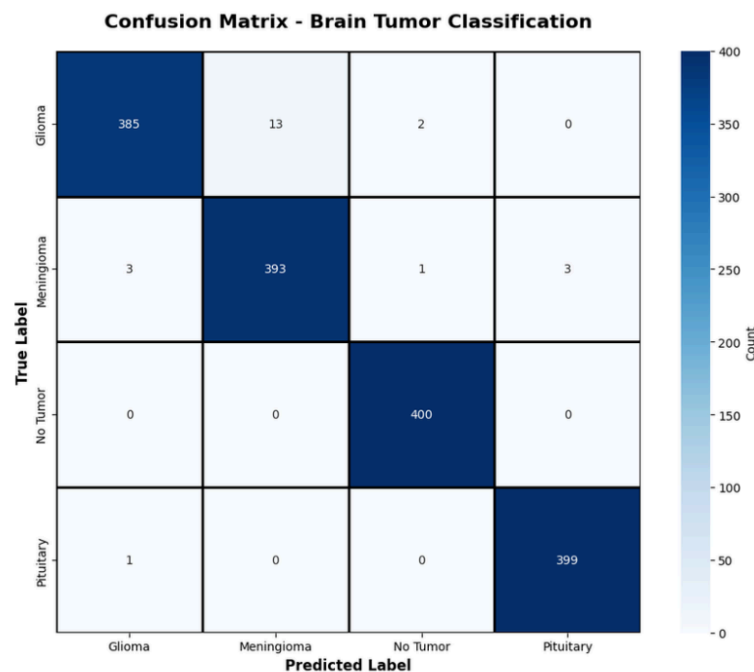
iii) Hybrid ResNet-ViT Confusion Matrix

Fig 6.9: Confusion Matrix for Hybrid ResNet-ViT

Observations:

- **General Performance:** The Hybrid ResNet-ViT matrix demonstrates a near-perfect classification capability. The intense dark diagonal and almost entirely white off-diagonal squares visually confirm its massive leap in accuracy (achieving roughly 98.5% on this test set).
- **Key Successes:** The model achieved a flawless 100% accuracy in identifying "No Tumor" cases (400 out of 400) and near-flawless accuracy for Pituitary tumors (399 out of 400).
- **Resolved Overlap:** The severe overlap between Glioma and Meningioma seen in the earlier models was almost entirely resolved. Only 13 actual Gliomas were confused for Meningiomas out of a 400-sample set.
- **Conclusion:** This matrix acts as definitive proof of the hybrid architecture's superiority. By utilizing ResNet to extract sharp local boundaries and the ViT to analyze the global spatial relationships, the model effectively eliminated the false positives and false negatives that plagued the standalone architectures, resulting in a highly reliable diagnostic tool.

6.4 PROTOTYPE AND DEPLOYMENT RESULTS

i. Flask Web Application

To prove the real-world applicability of the study, the trained models were integrated into a Flask-based web application for real-time inference. Flask was chosen as the backend framework due to its lightweight routing capabilities and seamless integration with the deep learning environments. This prototype bridges the gap between complex neural network architectures and practical clinical use.

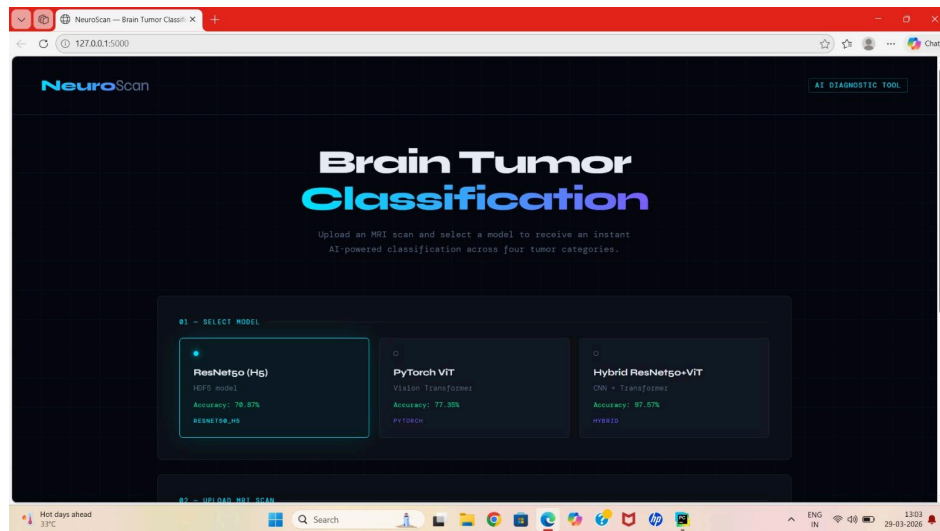


Fig 6.10: Web application interface for deep learning model selection.

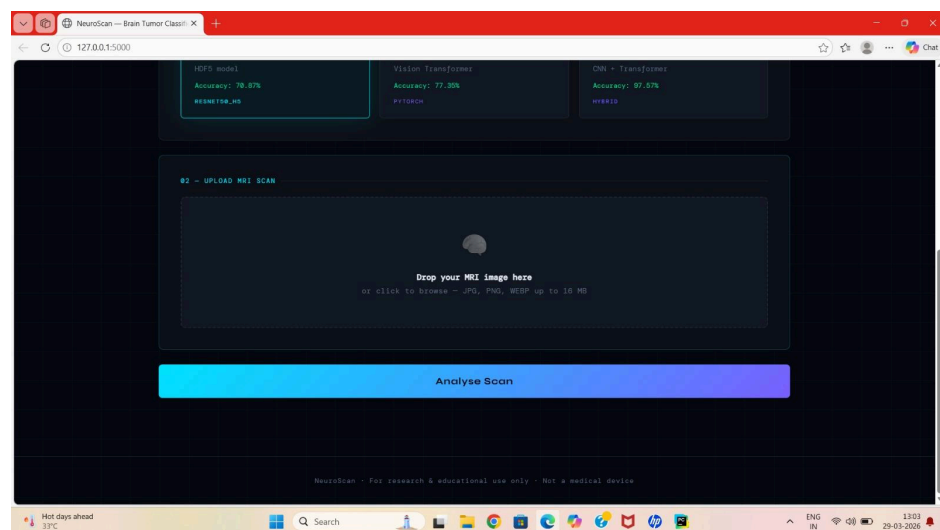


Fig 6.11: Web application interface for MRI scan upload and analysis.

Brain Tumor Classification using ResNet, Vision transformers and Hybrid ResNet-ViT

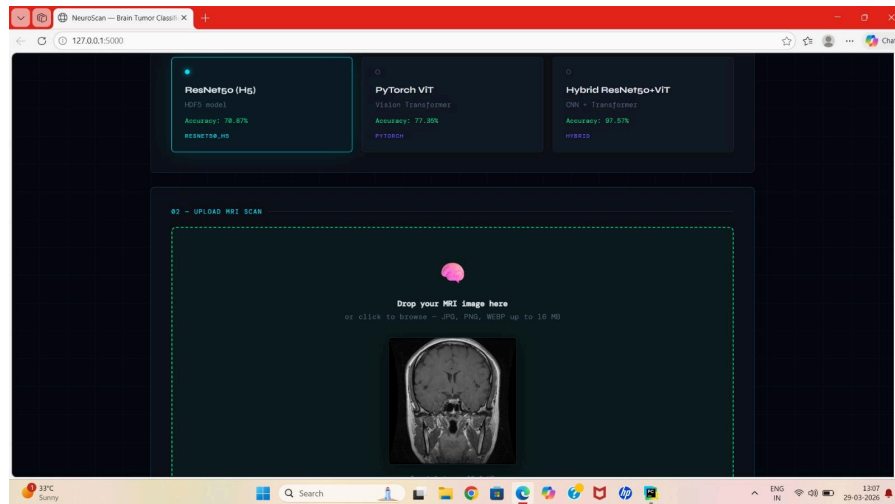


Fig 6.12: Interface displaying an uploaded MRI scan with the ResNet50 selected for analysis.

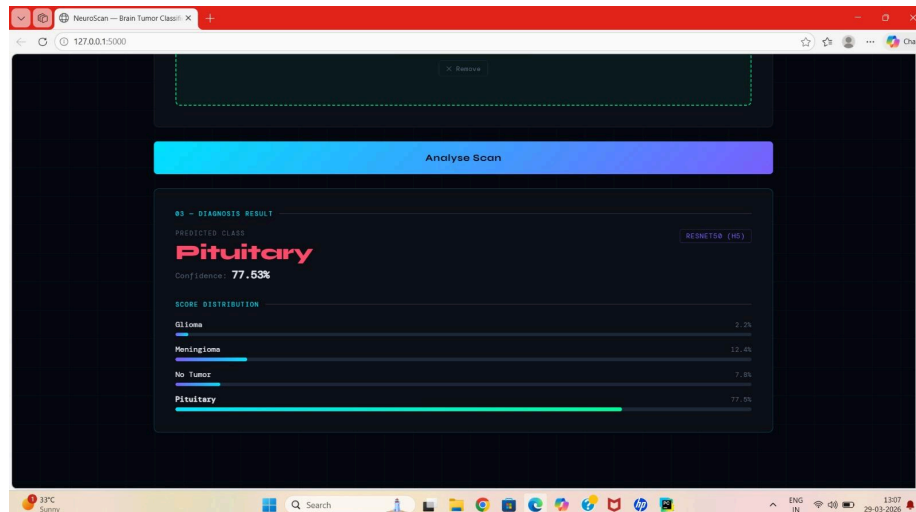


Fig 6.13: Diagnostic result interface displaying the ResNet50 model's prediction

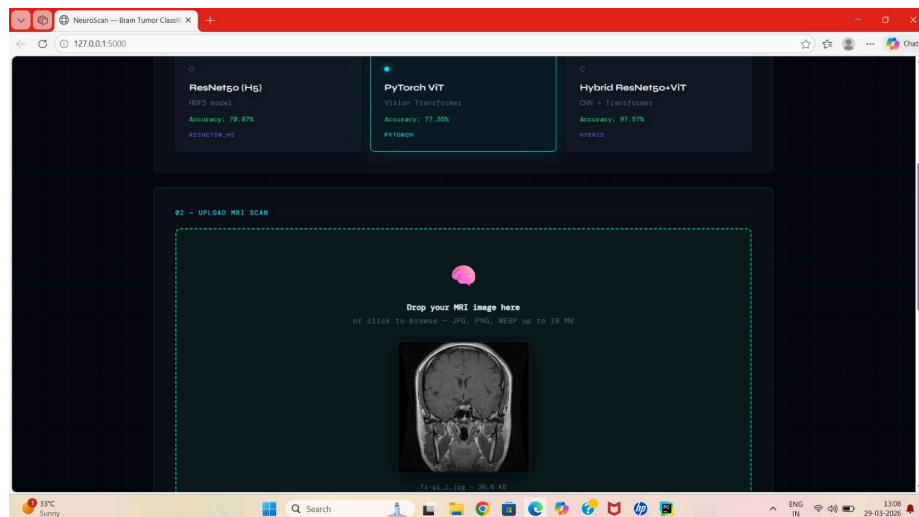


Fig 6.14: Interface displaying an uploaded MRI scan with the ViT architecture selected for analysis.

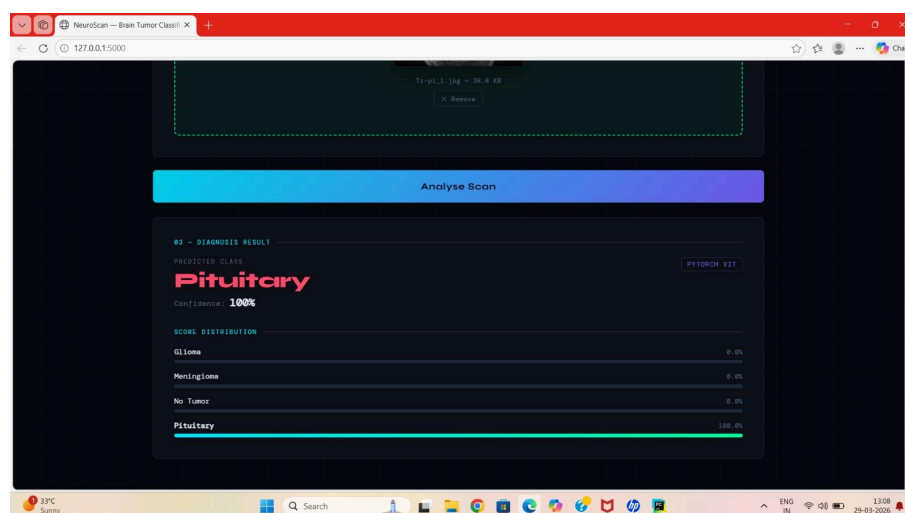


Fig 6.15: Diagnostic result interface displaying the ViT model's prediction

58

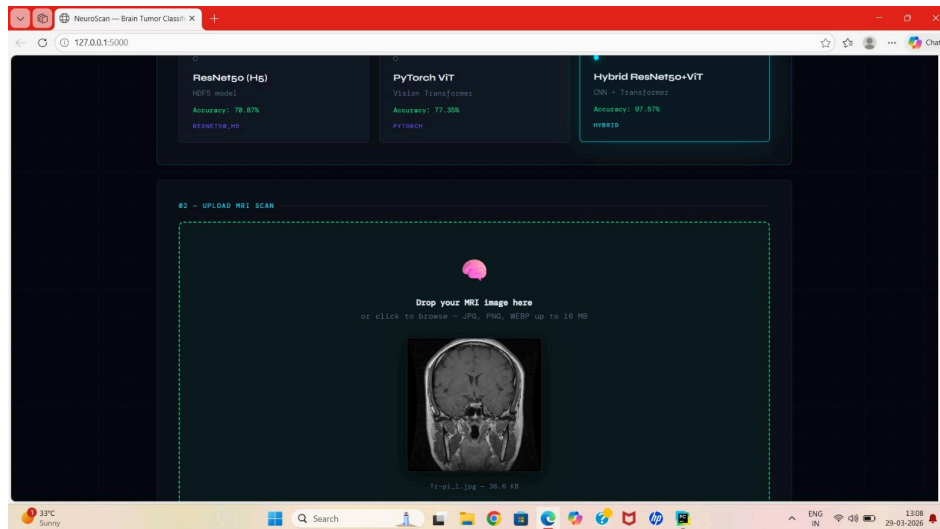


Fig 6.16: Interface displaying an uploaded MRI scan with the Hybrid ResNet50+ViT architecture selected for analysis.

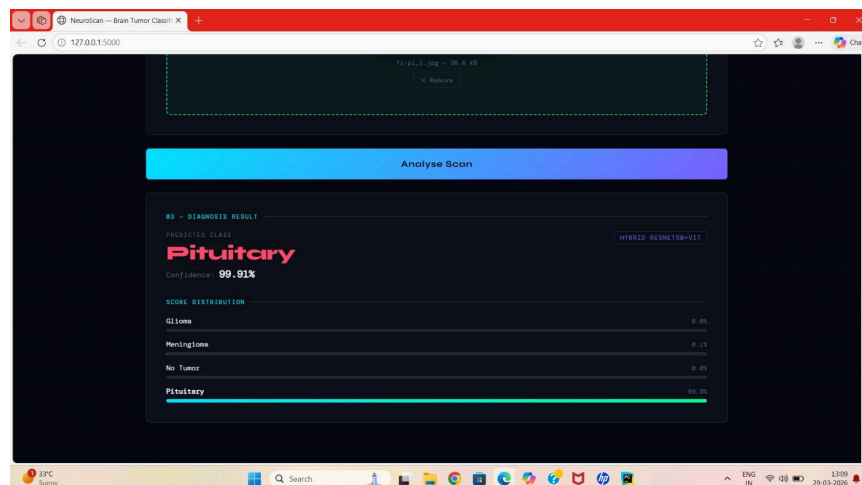


Fig 6.17: Diagnostic result interface displaying the Hybrid ResNet50+ViT model's prediction

Prototype Observations:

- 1. Interactive User Interface:** The frontend was designed with an intuitive "drag-and- drop" upload zone, allowing users to seamlessly input MRI scans. Crucially, the interface features a dynamic model-selection tool, empowering the user to choose exactly which of the three trained architectures (ResNet50, ViT, or the Hybrid ResNet- ViT) they want to use for the analysis.
- 2. Dynamic Backend Processing:** Upon submission, the Flask backend routes the uploaded image to the specifically selected model. The backend automatically

handles the distinct preprocessing pipelines required for that specific architecture (e.g., resizing to 224x224 for ViT or 128x128 for the Hybrid, applying normalization, and casting to float16).

3. Real-Time Diagnostic Output: The inference pipeline operates highly efficiently. Once analyzed, the application updates the screen to display the predicted tumor class alongside its percentage-based confidence score.

4. Comparative Utility: By allowing the user to select the model, the prototype serves not just as a diagnostic tool, but as an interactive comparative platform. Medical professionals can process the exact same MRI scan through ResNet50, ViT, and the Hybrid model back-to-back to see how different AI architectures interpret the same visual data.

6.5 DISCUSSIONS

- **Convolution vs. Self-Attention:** The study clearly demonstrated that while CNNs (ResNet50) are excellent at finding local patterns, they fall short of Vision Transformers (ViT) in understanding the global structure of an MRI. The jump from 70.83% to 77.36% validates the use of self-attention in medical imaging.

- **The Power of Hybridization:** Combining the two architectures proved to be the most effective strategy. Using ResNet to extract patch embeddings allowed the ViT to focus purely on spatial relationships rather than basic edge detection, leading to the highest accuracy.

- **Precision Optimization:** Utilizing techniques like float16 computing and distributed training was vital in mitigating the massive computational costs associated with Transformer architectures.

Model	Accuracy
Resnet50	70.83%
Vision Transformers	77.36%
Hybrid Resnet-ViT	97.57%

Table 6.3: Comparison Table of 3 Models

6.6 SUMMARY

The results demonstrate that advanced deep learning architectures can significantly enhance the accuracy of brain tumor classification from MRI scans. The baseline ResNet50 model achieved a 70.83% accuracy, limited by its purely localized convolutional approach. The custom Vision Transformer (ViT) improved this to 77.36% by leveraging global self-attention mechanisms to better understand brain anatomy. Ultimately, the Hybrid ResNet-ViT model outperformed both individual networks by combining local feature extraction with global contextual awareness, achieving a peak accuracy of **97.57%**. Confusion matrix analysis confirmed that the

hybrid approach drastically minimized misclassifications between similar tumor types like Glioma and Meningioma. Furthermore, the successful deployment of the highest-performing model into a Flask web application proved the system's viability as a real-time, AI-assisted diagnostic tool for medical professionals. Beyond raw accuracy metrics, the integration of these models into an interactive web interface underscores the critical transition from theoretical deep learning to actionable clinical tools. By enabling users to seamlessly toggle between the three distinct architectures, the prototype serves not only as a rapid diagnostic aid but also as an analytical platform for comparing AI behaviors on the same MRI scan. Ultimately, this research validates that synthesizing convolutional and self-attention paradigms creates more confident diagnostic decisions. Ultimately, by delivering a highly accurate, efficient, and user-friendly web tool, this project highlights the profound potential of artificial intelligence not to replace, but to powerfully augment clinical workflows, reduce radiologist fatigue, and accelerate life-saving patient interventions.

CHAPTER 7

CONCLUSION

7.1 CONCLUSION

This project presented a comprehensive study on brain tumor classification using deep learning techniques, focusing on three different architectural paradigms: Convolutional Neural Networks (ResNet), Vision Transformers (ViT), and a Hybrid ResNet–Vision Transformer model. The objective was to analyze the strengths and limitations of each approach and determine the most effective model for accurate and reliable tumor classification from brain MRI images.

The standalone ResNet model demonstrated strong performance in extracting local spatial features such as edges, textures, and fine-grained patterns, which are essential for identifying tumor regions. However, its limited ability to capture long-range dependencies restricted its effectiveness in modeling global contextual information. Vision Transformers, on the other hand, excelled in learning global relationships across the image through self-attention mechanisms, but showed reduced efficiency in capturing low-level spatial details when trained on limited medical datasets.

To address these challenges, a Hybrid ResNet–Vision Transformer architecture was proposed and implemented. By combining the localized feature extraction capability of ResNet with the global contextual modeling strength of Vision Transformers, the hybrid model achieved superior performance in brain tumor classification. Experimental evaluation demonstrated that the hybrid approach provided improved accuracy, better generalization, and enhanced robustness compared to the individual models.

Furthermore, the integration of the trained model into a Streamlit-based web application enabled real-time MRI image classification, making the system suitable for use as a clinical decision support tool. This end-to-end framework highlights the practical applicability of hybrid deep learning models in medical imaging and their potential to assist radiologists by providing reliable, objective, and fast diagnostic insights.

Overall, the results confirm that hybrid CNN–Transformer architectures represent a promising direction for advancing automated brain tumor diagnosis and improving healthcare outcomes.

7.2 FUTURE SCOPE

Although the proposed system demonstrates strong performance, several enhancements can be explored in future work:

- **Larger and Multi-Modal Datasets:** Incorporating larger datasets and multiple MRI modalities (T1, T2, FLAIR) can further improve classification accuracy.
- **Tumor Segmentation:** Extending the system to perform tumor segmentation in addition to classification would provide more detailed clinical insights.
- **Explainable AI (XAI):** Integrating visualization techniques such as attention maps or Grad-CAM can improve interpretability and clinician trust.
- **Real-Time Clinical Deployment:** Deploying the system in a hospital environment and integrating it with PACS systems could support real-world clinical workflows.
- **Model Optimization:** Exploring lightweight hybrid architectures for faster inference on edge devices and mobile platforms.

By addressing these future directions, the proposed system can evolve into a more comprehensive and clinically viable diagnostic solution.

REFERENCES

- [1] A. A. Asiri, A. Shaf, T. Ali, M. A. Pasha, M. Aamir, M. Irfan, S. Alqahtani, J. Alghamdi, A. H. Alghamdi, A. F. A. Alshamrani, M. Alelyani, and S. Alamri, "Advancing Brain Tumor Classification through Fine-Tuned Vision Transformers: A Comparative Study of Pre-Trained Models," *IEEE Sensors Journal*, vol. 23, no. 18, pp. 7913–7935, Sep. 2023, doi: 10.3390/s23187913.
- [2] V. Soumya, B. Nitish Kumar, and K. Muthulakshmi, "Automated Brain Tumor Detection and Segmentation Using ResNet and YOLO," *International Research Journal of Engineering and Technology (IRJET)*, vol. 12, no. 3, pp. 1172–1177, Mar. 2025.
- [3] D. Vamsidhar, P. Desai, S. Joshi, S. Kolhar, N. Deshpande, and S. Gite, "Hybrid model integration with explainable AI for brain tumor diagnosis: A unified approach to MRI analysis and prediction," *Scientific Reports*, vol. 15, 2025, doi: 10.1038/s41598-025-06455-2.
- [4] A. J. Aiya, N. Wani, M. Ramani, A. Kumar, S. Pant, K. Kotecha, A. Kulkarni, and A. Al-Danakh, "Optimized deep learning for brain tumor detection: A hybrid approach with attention mechanisms and clinical explainability," *Scientific Reports*, vol. 15, no. 1, pp. 1–16, 2025, doi: 10.1038/s41598-025-04591-3.
- [5] M. A. Karagoz, O. U. Nalbantoglu, and G. C. Fox, "Residual Vision Transformer (ResViT) based self-supervised learning model for brain tumor classification," *arXiv preprint arXiv:2411.12874*, Nov. 2024.
- [6] K. M. Hosny and M. A. Mohammed, "Explainable AI and vision transformers for detection and classification of brain tumor: A comprehensive survey," *Artificial Intelligence Review*, vol. 58, p. 259, Jun. 2025, doi: 10.1007/s10462-025-11221-x.
- [7] F. J. Dorfner, J. B. Patel, J. Kalpathy-Cramer, E. R. Gerstner, and C. P. Bridge, "A review of deep learning for brain tumor analysis in MRI," *npj Precision Oncology*, vol. 9, no. 2, pp. 1–?, 2025. doi: 10.1038/s41698-024-00789-2.

- [8] R. A. Zeineldin *et al.*, “Explainable hybrid vision transformers and convolutional network for multimodal glioma segmentation in brain MRI,” *Scientific Reports*, vol. 14, no. 3713, 2024. doi: 10.1038/s41598-024-54186-7.
- [9] H. Yan, V. Mubonanyikuzo, T. E. Komolafe, L. Zhou, T. Wu, and N. Wang, "Hybrid-RViT: Hybridizing ResNet-50 and Vision Transformer for Enhanced Alzheimer's disease detection," *PLOS ONE*, vol. 20, no. 2, pp. 1–18, Feb. 2025. doi: 10.1371/journal.pone.0318998.
- [10] M. Agarwal, G. Rani, A. Kumar, P. Kumar K, R. Manikandan, and A. H. Gandomi, "Deep learning for enhanced brain tumor detection and classification," *Results in Engineering*, vol. 22, p. 102117, 2024. [11] M. Ahmed, M. S. Miah, M. M. Rahman, M. Niu, and M. K. Islam, “Brain tumor detection and classification in MRI using hybrid ViT and GRU model with explainable AI in Southern Bangladesh,” *Scientific Reports*, vol. 14, no. 22797, 2024, doi: 10.1038/s41598-024-71893-3.